Australian Red Cross Lifeblood®

Blood component information (BCI)

An extension of blood component labels

2025 edition

Contents

Overview **2**

Blood component therapy 3
Donor selection, testing, and manufacturing 4
Blood component labelling 6
Anticoagulants and additive solutions 8
Storage, transport, and handling 9
Administration of blood components 10
Hospital transfusion committees 12

Component infomation and specifications **13**

Red cells 14
Paediatric red cells leucocyte depleted 15
Washed red cells leucocyte depleted 16

Apheresis platelets 17
Paediatric platelets 19
Platelets pooled 20

Fresh frozen plasma 21
Paediatric fresh frozen plasma 22
Cryodepleted plasma 23
Apheresis cryodepleted plasma 24
Cryoprecipitate 25
Apheresis cryoprecipitate 26

Appendices **27**

Appendix I: Patient Blood Management Guidelines 28
Appendix II: Clinical indications for blood components that have been further manufactured or tested 29
Appendix III: Blood component modifications 31
Appendix IV: Transfusion reactions 32
Appendix V: Residual risk estimates for transfusion-transmitted infections 41

References **42**



Overview

Lifeblood's critical role in health care is the provision of a safe, secure and cost-effective supply of quality blood products.

This resource is provided in accordance with the regulations of the Therapeutic Goods Administration (TGA) and is considered an extension of our blood component labels due to limited space on these labels.

The BCI provides details for each blood component including the blood collection process, method of manufacture, critical manufacturing steps, clinical indications for use, and administration methods.





Blood component therapy

Blood and blood components are biological products, and in the form of cellular products are living human tissue intended for use in the treatment of patients.

Transfusion can effectively and efficiently provide or replace missing or malfunctioning elements of the blood or immune systems, both in the short and long term.

Professional judgement based on clinical evaluation determines the selection of components, the dosage, and the rate of administration.

Principles of blood component therapy

- The decision to transfuse should be based, wherever possible, on an appropriate clinical and laboratory diagnosis.
- The decision to transfuse must take into account the risks and benefits to the patient.
- Before making the decision to transfuse, consideration should be given to availability, cost and the importance of appropriately using a valuable, freely given community resource.

Selection of blood components

Patients receive specific blood components manufactured from whole blood or apheresis donations.

As with other medicines, each blood component has specific clinical indications, known benefits, and potential risks.

The need for transfusion should take into account the patient's unique circumstances, and the chosen component(s) and dose should be tailored to their clinical situation.

Patient blood management (PBM)

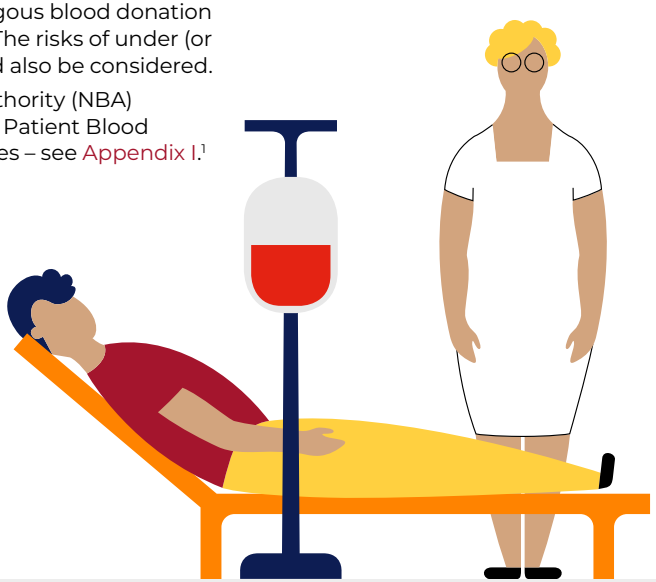
To prevent inappropriate transfusions, clinicians are encouraged to adopt the principles of PBM and prescribe blood components based on appropriate clinical indications and guidelines. PBM is the timely application of evidence-based medical and surgical practices designed to maintain haemoglobin concentration, optimise haemostasis and minimise blood loss. As a consequence of better management, patients usually require fewer transfusions thus reducing the risk of transfusion-associated complications.

Alternatives to blood transfusion including iron therapy or blood conservation techniques, such as intra-operative cell salvage, should be considered when appropriate, to reduce the potential risks of disease transmission and immune reactions. It should be noted, however, that routine use of preoperative autologous blood donation is not recommended. The risks of under (or not) transfusing should also be considered.

The National Blood Authority (NBA) has published national Patient Blood Management Guidelines – see Appendix I.[1]

Risks and adverse events

Careful donor selection and laboratory testing significantly reduces but does not totally eliminate all potential hazards of blood transfusion. There remains risk of transmitting infectious agents, including bacteria, viruses, parasites. In addition, blood components may contain immunising agents other than those indicated on the label. For example, a unit of platelets also contains residual red blood cells and white blood cells. Serious transfusion reactions, although rare, may be life-threatening.



Donor selection, testing, and manufacturing

Blood is collected from voluntary, non-remunerated donors within guidelines based on recommendations from the World Health Organization (WHO), International Society of Blood Transfusion (ISBT), and the International Federation of Red Cross and Red Crescent Societies.

Donor selection and collection

Donors attend permanent and mobile collection centres across Australia.

Blood is collected either as a whole blood donation or by apheresis for donations of plasma or platelets. Prior to their donation, donors will have:

- satisfactorily completed a confidential interview and donor declaration regarding high-risk behaviour, practices and circumstances which would prevent them from donating
- satisfactorily completed a health assessment that includes a questionnaire on past and present medical conditions
- satisfied minimum physiological criteria, and
- been told to contact Lifeblood, including after their donation, if they have any health issues or other information which may affect the suitability of their donation.

In identifying suitable donors and accepting their blood donations, ensuring the safety of both the donor and the eventual transfusion recipient is paramount.

In exceptional circumstances, Lifeblood will also collect autologous or directed donations.

Testing of donor blood

All blood donations are tested in order to enable appropriate and safe selection of blood for transfusion, for example, the importance of correctly determining the donor's blood group in ensuring ABO-compatible transfusions, and prevent (where possible) transfusion transmission of infectious agents which can cause diseases in transfusion recipients.

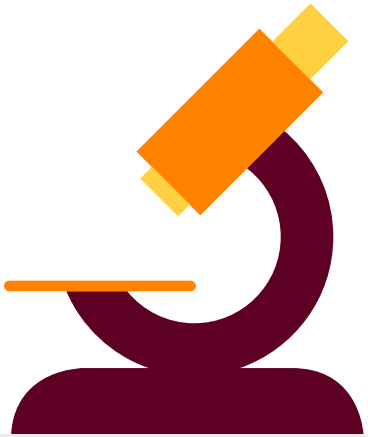
In Australia, the following mandatory tests are performed on all blood donations used for fresh components: blood group (ABO and RhD), red cell antibody screen and infectious disease screening for human immunodeficiency virus (HIV) 1 and 2, hepatitis B virus (HBV), hepatitis C virus (HCV) and syphilis. New donors, and donors of leucocyte components, are also tested for human T-cell leukaemia virus (HTLV) types 1 and 2.

All mandatory testing is performed in TGA licensed Lifeblood facilities, according to the principles of good manufacturing and laboratory practice (GMP/GLP), and following the manufacturers' instructions and strict Lifeblood guidelines and standard operating procedures.

Only donations that pass mandatory testing and meet the necessary specifications are released for clinical use or further processing. If an infectious disease screening test is confirmed positive, the donation is destroyed.

Bacterial contamination of platelets is recognised as the most significant residual infectious risk of transfusion in developed countries. In order to minimise this risk, Lifeblood screens all platelet components for bacterial contamination. Donations collected exclusively for plasma fractionation undergo infectious diseases screening for HIV, HBV and HCV but are not tested for HTLV 1/2, syphilis or a blood group, and an antibody screen is only performed if the donor was transfused or pregnant since last donating.

CSL Behring engaged an external laboratory to perform hepatitis A and parvovirus B19 nucleic acid testing (NAT) on plasma supplied by Lifeblood, for plasma quality assurance release testing, from April 2025. All donations undergo this additional testing. Hepatitis A and parvovirus B19 are not release tests for fresh component transfusion. Lifeblood is notified of positive results and performs product recall and clinician notification where the component has been transfused.



Donor selection, testing, and manufacturing continued

Processing blood into components

Blood components are an effective and efficient use of donated whole blood and other resources. Component therapy means patients can receive treatment tailored to their particular clinical situation or condition.

Blood components are prepared from whole blood donations using various methods of physical separation, such as centrifugation. Lifeblood uses centralised processing facilities where standardised procedures, based on a pharmaceutical production model and good manufacturing and laboratory practice (GMP/GLP), support the controlled and cost-effective production of high quality, consistent and safe blood components.

Our regulatory requirements

The manufacturing activities of Lifeblood are regulated by the Therapeutic Goods Administration under the *Therapeutic Goods Act 1989* and the following applicable standards:

- *Therapeutic Goods (Manufacturing Principles) Determination 2020*
- *Australian Code of Good Manufacturing Practice for Human Blood and Blood Components, Human Tissues and Human Cellular Therapy Products* Version 1.0 May 2013 (cGMP)
- Council of Europe – *Guide to the Preparation, Use and Quality Assurance of Blood Components* (CoE Guide) 19th edition
- *Therapeutic Goods Order No. 102 – Standard for Blood and Blood Components* (TGO 102)
- *Therapeutic Goods Order No. 108 - Standard for Human Cell and Tissue Products - Donor Screening Requirements* (TGO 108)



Blood component labelling

Current ISBT 128 transition labelling

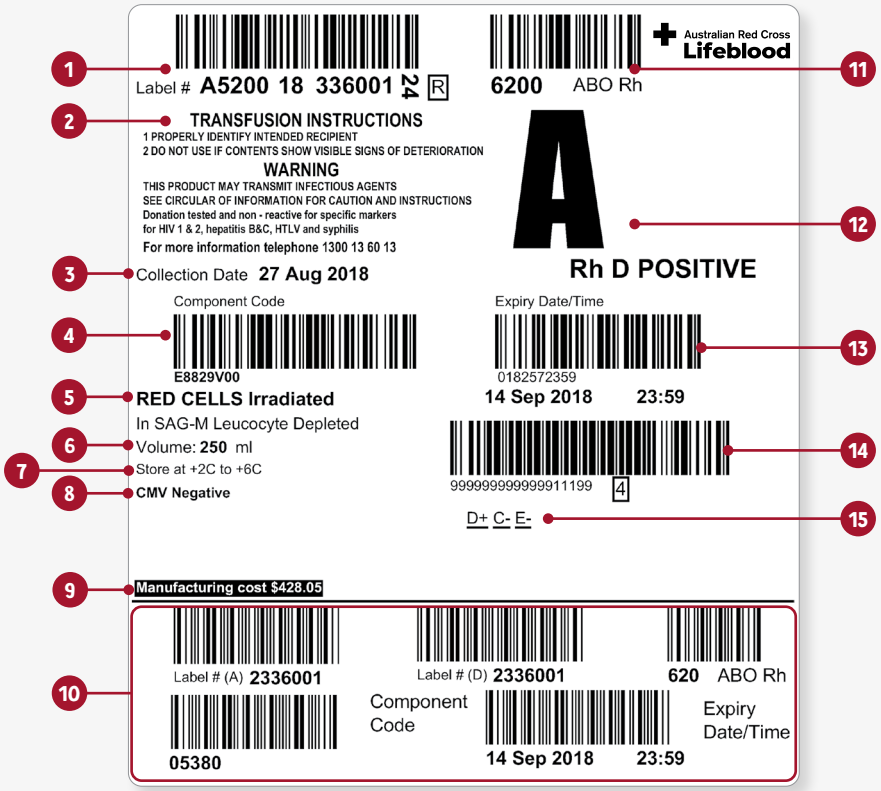
ISBT 128 is a global standard for the identification, labelling and information transfer of Medical Products of Human Origin (MPHO). It aims to achieve international consistency in the information provided on MPHO labels through global registration of product codes through the International Council for Commonality in Blood Banking Automation (ICCBBA).

The standard has been adopted by hospitals, blood centres, tissue and cellular therapy facilities and plasma fractionators in over 75 countries. Adopting this standard provides a unique identifier for Australian blood components and will prevent worldwide duplication of donation identification numbers within a 100-year period.

To facilitate the changeover to ISBT 128 labelling, which is planned to occur on 30 November 2025, Lifeblood is using a 'transition label' for its blood components. This allows customers who are not yet 'ISBT 128 capable' to continue managing their Codabar labelled inventory without disruption. The upper portion of the transition label has ISBT 128 barcodes and the lower portion has a distinct transition zone that provides the existing Codabar barcodes. When components with clinically important special attributes such as irradiation, modifiers, and additional testing are required, only those where the desired characteristic is confirmed on the label should be selected. Modifier texts and their explanations is provided in Appendix III.

Further information on ISBT 128 labelling and data structures, including *The Australian Guidelines for the Labelling of Blood Components using ISBT 128*, can be found at lifeblood.com.au

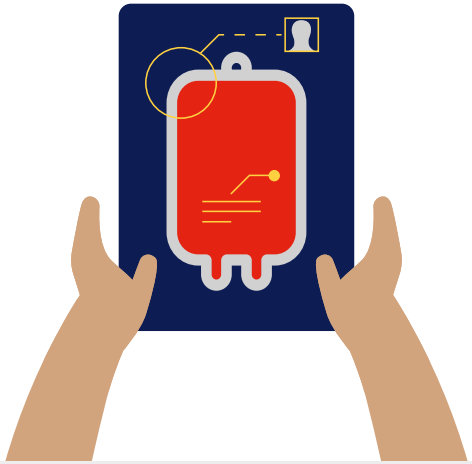
1. Donor identification number of the component (barcode and eye-readable), or pool number in the case of a pooled component
2. Transfusion instructions, warning and infectious diseases statement
3. Collection date (eye-readable) of the donation from which the component was made, or the preparation date of a pooled component
4. Component code (barcode and eye-readable.
5. Component proper name and other information e.g. anticoagulants or additives
6. Contents or volume
7. Storage conditions
8. Component modifiers, or special attributes that are not part of the component code
9. Indicative manufacturing cost
10. Transition zone with Codabar barcodes The eye-readable 7-digit Donation Identification Number for both Label #A and Label #D is identical. If scanning, Label #A barcode has "A" start and stop codes and Label #D barcode has "D" start and stop codes.
11. ABO and RhD blood group code (barcode and eye-readable)
12. ABO and RhD blood group (eye-readable)
13. Expiry date and time (barcode and eye-readable)
14. Special testing barcode unique to the ISBT 128 label
15. Special test result area (eye-readable) e.g. red cell antigens or platelet HLA or platelet-specific antigens



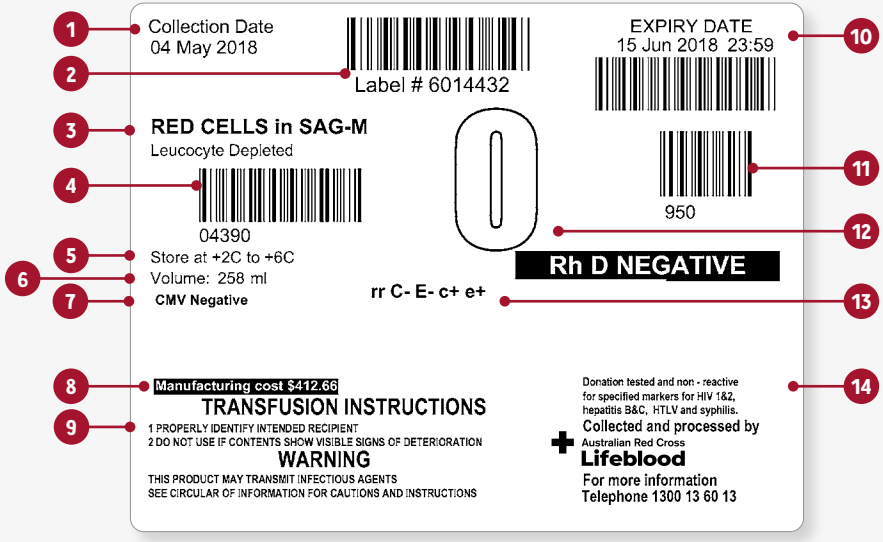
Blood component labelling continued

Legacy Codabar labelling

Frozen red cells with extended expiry collected prior to the implementation of ISBT 128 labelling will continue to be labelled using the original Codabar standard. Further information about Codabar labels can be found on our website at lifeblood.com.au



1. Collection date (eye-readable) of the donation from which the component was made, or the preparation date of a pooled component
2. Donor identification number of the component (barcode and eye-readable), or pool number in the case of a pooled component
3. Component description
4. Component code (barcode and eye-readable) and proper name and other information e.g. anticoagulants or additives
5. Storage conditions
6. Contents or volume
7. Component modifiers, or special attributes that are not part of the component code
8. Indicative manufacturing cost
9. Transfusion instructions, warning and infectious diseases statement
10. Expiry date and time (barcode and eye-readable)
11. ABO and RhD blood group code (barcode and eye-readable)
12. ABO and RhD blood group (eye-readable)
13. Special test result area (eye-readable) e.g. red cell antigens or platelet HLA or platelet-specific antigens
14. Infectious diseases statement



Anticoagulants

Whole blood collections	Constituents	g/L	Total per pack (g)
CPD (citrate phosphate dextrose) Macopharma 66.5 mL per pack of whole blood	Sodium citrate dihydrate	26.3	1.75
	Citric acid monohydrate	3.27	0.22
	Monobasic sodium phosphate dihydrate	2.51	0.17
	Dextrose monohydrate	25.5	1.70

Platelet apheresis collections	Constituents	g/L
ACD-A (acid citrate dextrose) Terumo BCT 750 mL Mixed ~ 1:11 with whole blood	Dextrose monohydrate	24.5
	Sodium citrate dihydrate	22.0
	Citric acid monohydrate	8

Plasma apheresis collections	Constituents	g/L
4% sodium citrate Haemonetics Corporation 250 mL mixed ~ 1:16 with whole blood	Sodium citrate dihydrate	40



Additive solutions

Red cell additive solution	Constituents	g/L	Total per pack (g)
SAG-M (saline adenine glucose mannitol) Macopharma 105 mL of SAG-M is used as the additive solution for resuspension of standard red cell components.	Sodium chloride	8.77	0.92
	Adenine	0.169	0.02
	Glucose monohydrate	9.0	0.95
	Mannitol	5.25	0.55

Red cell additive solution (washed red cells, red cells for IUT and thawed frozen red cells)	Constituents	g/L	Total per pack (g)
SAG-M (saline adenine glucose mannitol) Macopharma 100 mL of SAG-M is used as the additive solution for resuspension of washed red cells (after washing with 250 mL SAG-M), red cells for IUT and thawed frozen red cells.	Sodium chloride	8.77	0.877
	Adenine	0.169	0.0169
	Glucose monohydrate	9.0	0.9
	Mannitol	5.25	0.525

Platelet additive solution (PAS)	Constituents	mg/100 mL
SSP+* Macopharma The volume of platelet additive solution contained in each unit of platelets pooled in platelet additive solution leucocyte depleted is approximately 70% of the total component volume. The volume of platelet additive solution contained in each unit of platelets apheresis in platelet additive solution leucocyte depleted is approximately 60% of the total component volume.	Sodium chloride	405
	Sodium acetate trihydrate	442
	Sodium citrate dihydrate	318
	Sodium dihydrogenophosphate	105
	Di-sodium hydrogenophosphate	305
	Potassium chloride	37
	Magnesium chloride	30

**Note: PAS-E is the generic nomenclature adopted by ICCBBA to describe platelet additive solutions in terms of their active ingredients: citrate, phosphate, acetate, magnesium and potassium. This nomenclature allows for solutions with the same active ingredients from different commercial sources to be coded in the same manner. SSP+ is one of the PAS-E Platelet Additive Solutions.*

Storage, transport, and handling

Blood components must be stored and transported at controlled temperatures in accordance with the requirements shown in the table.[2,3]

All components

Keep blood components in an appropriately monitored temperature-controlled environment until administered.

Transfuse blood components as soon as possible after removal from temperature-controlled storage.

Minimise the amount of time blood components are outside of temperature-controlled storage to ensure that the specified temperature limits are not exceeded.

Handling and storage must minimise the possibility of product tampering.

Red cells

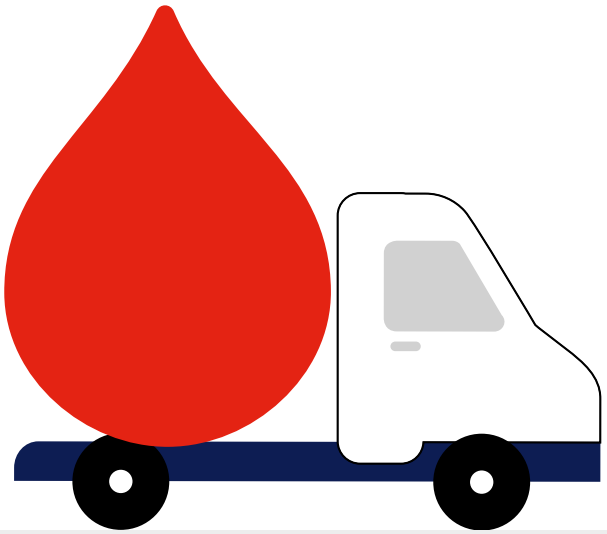
On each occasion, the amount of time red cell components are outside of temperature-controlled storage but not transfused should be restricted to less than 30 minutes.

Frozen components

Thaw frozen blood components using an approved method such as a temperature-controlled waterbath maintained between 30°C and 37°C or in an approved microwave device.

Prevent water contamination of entry ports during thawing, for example, using a watertight protective plastic over-wrap. Do not thaw frozen components in a domestic microwave oven or using hot water directly from a tap.

Component	Storage temperature	Transport temperature	Comments
Red cells	2°C to 6°C	2°C to 10°C	Blood refrigerators including those in theatre and other wards or remote locations must be managed in accordance with AS3864-Part 2 (2012).[4]
Platelets	20°C to 24°C	20°C to 24°C	Must be agitated gently and continuously in a single layer on a platelet agitator.
Fresh frozen plasma **Cryoprecipitate** **Cryodepleted plasma**	–25°C or below	–25°C or below	Freezers must be managed in accordance with AS3864-Part 2 (2012).[4]



Administration of blood components

Lifeblood recommends that transfusing facilities issue guidelines and make them widely available with the support of their clinical, scientific, and other key staff.

All clinical staff involved in ordering, preparing, issuing, and transfusing blood components must be trained in the correct procedures, and must be familiar with the general and institutional requirements for transfusion practice.

Resources for the creation of guidelines

In developing institutional guidelines, there are a wide range of resources including:

- National Blood Authority Patient Blood Management Guidelines[1]
- Australian Commission on Safety and Quality in Health Care (ACSQHC) Blood Management Standard[5]
- Australian and New Zealand Society of Blood Transfusion Guidelines for Transfusion and Immunohaematology Laboratory Practice[2]
- Australian and New Zealand Society of Blood Transfusion and Australian College of Nursing Guidelines for the Administration of Blood Products[3]
- Australian Red Cross Lifeblood. Blood Book: Australian Blood Administration Handbook[6]
- National Pathology Accreditation Advisory Council (NPAAC) Requirements for Transfusion Laboratory Practice[7]
- World Health Organization Educational Modules on Clinical Use of Blood,[8] and
- British Society for Haematology Guideline on the Administration of Blood Components.[9]

The ANZSBT guidelines and NPAAC requirements provide guidance especially in the areas of sample identification, compatibility testing, issue and transfusion of blood components, and investigation of transfusion reactions.



Administration of blood components

Patient identification

To ensure the correct blood component is administered, the intended recipient must be properly identified at each stage of the transfusion process, from collection and labelling of the sample and request form through to commencement of the transfusion.

When a blood component is received for transfusion, the patient, prescription and pack details must be confirmed according to approved procedures. The checking procedure must be performed independently by two appropriately qualified staff, with each independently carrying out and taking responsibility for the checking process.

Documentation

The decision to transfuse must be clearly documented in the patient's medical record and on the request form, and include the prescribed component, clinical indication for transfusion, number of packs (or volume/dose required) and special requirements. The identity and signatures of staff members performing the identity checks and administering the transfusion must be recorded. The unique donor identification number (DIN) of each pack being transfused, the start and finish times, the patient's observations, as well as other relevant details, must be recorded in the patient's medical record. Records should be retained for as long as required by local policies and state/territory and national regulations.

Inspection of components

Whenever being handled, prior to issue from the laboratory and before transfusion, check the appearance of the blood component. Do not transfuse if there is any evidence of haemolysis, clot formation, significant colour change in the contents (e.g. in the red cells compared to crossmatch segments), tampering or any other reason that the blood component appears unsuitable for use. Return the affected blood component to the Transfusion Service Provider or Lifeblood for further evaluation.

Adverse transfusion reactions

Frequent visual observation of the patient before, during and after the transfusion is essential to identify possible adverse reactions. If an adverse transfusion reaction occurs, stop the transfusion immediately and initiate appropriate therapy. The transfusion should not be restarted unless the patient receives a satisfactory clinical review.

All significant adverse transfusion reactions, including possible bacterial contamination of a blood component, suspected disease transmission or transfusion-associated acute lung injury (TRALI) should be immediately reported to the Transfusion Service Provider. The remainder of any implicated blood components should be retained for further investigation. If an adverse reaction has blood component safety or quality implications and may require donor investigations and/or recall of associated blood components it should also be reported to Lifeblood.

For further information on adverse transfusion reactions see Appendix IV or lifeblood.com.au

Administration procedures

Transfusion of blood components should only be undertaken where appropriate staff and facilities are available to identify and manage potential adverse transfusion reactions. Local policies and procedures should always be followed. Detailed guidance is provided by the ANZSBT/ACN Guidelines for the Administration of Blood Products.[4]

- Blood components are prepared in a sterile closed system to minimise the risk of bacterial contamination. Once opened or 'spiked', the blood component should be transfused within four hours of opening if maintained at room temperature (20–24°C).
- Blood components should be transfused through an approved blood administration set incorporating a standard (170–200 µm) filter to remove particulate matter. Bedside leucocyte filters are not required because all cellular components are leucocyte depleted.
- Blood components should be thoroughly mixed by inversion before use.
- Blood components may be warmed just prior to or during transfusion, if clinically indicated. Only approved blood warming devices should be used.
- Only one type of blood component should be administered at a time, unless in an emergency situation. Multiple packs of the same type of blood component (e.g. red cells) can be administered consecutively through the same giving set, according to the manufacturer's recommendations.
- Medication must not be added to the blood component bag or blood giving set or intravenous (IV) line.
- IV fluids must not be co-administered with blood components unless there is sufficient data to ensure compatibility. The only IV fluid universally compatible with blood components is 0.9% sodium chloride. Red cells are compatible with ABO-compatible plasma and albumin.[3] Electrolyte and colloid solutions containing calcium (such as Hartmann's solution, lactated Ringer's solution and Haemaccel®) may cause clots to form in the blood component in the presence of citrate anticoagulant.
- Where clinically appropriate, the infusion should start slowly. The patient should be closely observed for the initial 15 minutes of the infusion as life-threatening reactions may occur after infusion of only a small volume. After 15 minutes, the rate of infusion may be increased to the maximum prescribed rate, provided there are no signs or symptoms of an adverse reaction.
- Transfusion of each pack should be completed prior to the labelled expiry or within four hours of removal from temperature-controlled storage, whichever is sooner.

Hospital transfusion committees

Hospital transfusion (or patient blood management) committees play an important role in monitoring adherence to national standards and guidelines, developing local institutional clinical transfusion practice policies and procedures, and supervising educational, training and audit activities and the reporting of adverse reactions.[10]

The National Safety and Quality Health Service: *Blood Management Standard* requires that: "*Leaders of a health service organisation describe, implement and monitor systems to ensure the safe, appropriate, efficient and effective care of patients' own blood, as well as other blood and blood products. The workforce uses the blood product safety systems.*"[5]

Guidance with respect to the establishment and responsibilities of hospital transfusion committees is provided within the:

- Australian and New Zealand Society of Blood Transfusion (ANZSBT) *Guidelines for Transfusion and Immunohaematology Laboratory Practice*[2]
- Australian and New Zealand Society of Blood Transfusion and Australian College of Nursing *Guidelines for the Administration of Blood Products*[3]



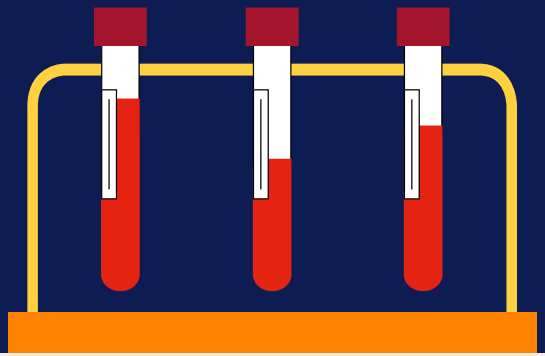


Component information and specifications

Blood components are manufactured to specifications mandated by the Council of Europe (COE) Guide to the Preparation, Use and Quality Assurance of Blood Components (19th Edition; 2017).[11]

Component quality and processing methods are monitored using statistical process control (SPC). The testing protocol takes into account all major production variables with sampling plans reflecting these.

The sampling rate is set to ensure statistical validity and as a general principle, guidelines require the monitoring of 1% (or 10 products per month, whichever is greater) although, for some parameters in specific component types, all packs will be tested. Lifeblood requires a minimum of 90% of components tested meet specifications.





Red cells
leucocyte depleted

Description	A red cell component obtained by removing most of the plasma after centrifuging whole blood collected into anticoagulant. The red cells may be resuspended in other additives (e.g. SAG-M) to prolong storage and are filtered to remove most leucocytes. May be irradiated.
Indications	For treatment of clinically significant anaemia with symptomatic deficit of oxygen-carrying capacity, and replacement of traumatic or surgical blood loss.
Contraindications	Depending upon the condition of the patient, transfusion of red cells may not be necessary even with low haemoglobin concentration. Do not use red cells if anaemia can be treated with specific therapies such as iron, vitamin B12, folic acid or erythropoietin and the clinical condition of the patient permits sufficient time for these to promote erythropoiesis.

Typical unit content

Data from:
1 January 2024 to
31 December 2024

Parameter	Mean (± 1 SD)	Specification
Volume (mL)	257 ± 15	> 220
Haemoglobin (g/unit)	49 ± 5	≥ 40
Haematocrit (L/L)	0.59 ± 0.03	0.50–0.70
Haemolysis (% at expiry)	0.3 ± 0.1	< 0.8
Leucocyte count (x10^6/unit)	0.03 ± 0.09	< 1.0

Availability	Available in group O, A, B and AB; and RhD positive and RhD negative.
Shelf life	42 days with the appropriate additives. Red cells may be irradiated at any time up to 14 days after collection, and thereafter post-irradiation shelf life is 14 days.
Storage temperature	2°C to 6°C. Time outside of controlled storage conditions prior to commencing transfusion should not exceed 30 minutes.[2,3,6]
Dosage	Each unit raises haemoglobin concentration by approximately 10 g/L. Transfuse one unit and reassess patient.
Administration	Transfuse through blood administration set incorporating a standard (170–200 µm) filter. Transfusion of each unit should be completed within four hours of removal from approved controlled storage.
Transfusion reactions	See Appendix IV: Transfusion reactions
Modifications	Phenotyped, CMV seronegative, irradiated. See Appendix II: Clinical indications for blood components which have been further manufactured or tested
Comments	Whenever possible, blood of identical ABO and RhD group to the recipient should be used. However, group O red cells must be used in an emergency when the recipient's blood group is unknown.[3] In this situation, a blood sample should be taken for pretransfusion testing prior to commencing transfusion.

Paediatric red cells
leucocyte depleted

Description	A leucocyte-depleted red cell component divided into four packs of equal volume for the purpose of reducing donor exposure for small paediatric transfusions and to minimise product wastage. May be irradiated.
Indications	For treatment of clinically significant anaemia with symptomatic deficit of oxygen-carrying capacity in infants and young children. May also be used for intrauterine transfusion.
Contraindications	Depending upon the condition of the patient, transfusion of red cells may not be necessary even with low haemoglobin concentration. Do not use red cells if anaemia can be treated with specific therapies such as iron, vitamin B12, folic acid or erythropoietin and the clinical condition of the patient permits sufficient time for these to promote erythropoiesis.

Typical unit content

Data from:
1 January 2024 to
31 December 2024

Parameter	Mean (± 1 SD)	Specification
Volume (mL)	59 ± 4	25–100
Haemoglobin (g/unit)	NA	Initial unit prior to splitting ≥ 40
Haematocrit (L/L)	0.63 ± 0.03	0.50–0.70
Haemolysis (% at expiry)	0.2 ± 0.1	< 0.8
Leucocyte count (x10^6/unit)	NA	Initial unit prior to splitting < 1.0

Availability	Must be requested in advance, unless by local arrangement.
Shelf life	35 days with the appropriate additives. Paediatric red cells may be irradiated at any time up to 14 days after collection, and thereafter post-irradiation shelf life is 48 hours.
Storage temperature	2°C to 6°C. Time outside of controlled storage conditions prior to commencing transfusion should not exceed 30 minutes.[2,3,6]
Dosage	In neonates and children < 20 kg, transfusion volume should be calculated based on weight and prescribed in mL. In children > 20 kg use one adult unit and reassess patient. Common paediatric dosing is 10–15 mL/kg, alternatively the following formula can be used: Volume required = weight (kg) x Hb rise required (g/L) x 0.5
Administration	Transfuse through blood administration set incorporating a standard (170–200 µm) filter. Transfusion of each unit should be completed within four hours of removal from approved controlled storage.
Transfusion reactions	See Appendix IV: Transfusion reactions
Modifications	Phenotyped, CMV seronegative, irradiated. See Appendix II: Clinical indications for blood components which have been further manufactured or tested
Comments	Whenever possible, blood of identical ABO and RhD group to the recipient should be used.

Washed red cells
leucocyte depleted

Description	Red cells are washed with sterile SAG-M (saline adenine glucose mannitol) solution using a manual process to remove the majority of unwanted plasma proteins, antibodies and electrolytes. The washed red cells are resuspended in SAG-M additive solution. May be irradiated.
Indications	For treatment of clinically significant anaemia with symptomatic deficit of oxygen-carrying capacity in infants and young children. May also be used for intrauterine transfusion. Indicated for patients requiring red cells with a low protein supernatant, such as those experiencing reactions to transfused plasma e.g. patients who have IgA deficiency and antibodies to anti-IgA. Washed red cells may reduce the incidence of severe recurrent febrile, urticarial and possible anaphylactic reactions in multi-transfused recipients. May also be considered for reducing isoagglutinins for transfusion post paediatric ABO-mismatched cardiac transplant.
Contraindications	Depending upon the condition of the patient, transfusion of red cells may not be necessary even with low haemoglobin concentration. Do not use red cells if anaemia can be treated with specific therapies such as iron, vitamin B12, folic acid or erythropoietin and the clinical condition of the patient permits sufficient time for these to promote erythropoiesis.

Typical unit content

Data from:
1 January 2024 to
31 December 2024

Parameter	Mean (± 1 SD)	Specification
Volume (mL)	261 ± 15	> 130
Haemoglobin (g/unit)	50 ± 5	≥ 40
Haematocrit (L/L)	0.63 ± 0.02	0.50–0.70
Haemolysis (% at expiry)	0.2 ± 0.1	< 0.8
Leucocyte count (x10^6/unit)	NA	Initial unit prior to splitting < 1.0
Last wash supernatant (total protein g/unit)	0.01 ± 0.00	< 0.5

Availability	Must be requested in advance.
Shelf life	28 days. Washed red cells may be irradiated at any time up to 14 days after collection, and thereafter post-irradiation shelf life is 14 days.
Storage temperature	2°C to 6°C. Time outside of controlled storage conditions prior to commencing transfusion should not exceed 30 minutes.[2,3,6]
Dosage	Each unit raises haemoglobin concentration by approximately 10 g/L. Transfuse one unit and reassess patient.
Administration	Transfuse through blood administration set incorporating a standard (170–200 µm) filter. Transfusion of each unit should be completed within four hours of removal from approved controlled storage.
Transfusion reactions	See Appendix IV: Transfusion reactions
Modifications	Phenotyped, CMV seronegative, irradiated. See Appendix II: Clinical indications for blood components which have been further manufactured or tested
Comments	As there is some loss of red cells during processing, the increment in haemoglobin from one unit of washed red cells is less than expected from an unwashed unit. Whenever possible, blood of identical ABO and RhD group to the recipient should be used.

Apheresis platelets
in platelet additive solution, leucocyte depleted

Description	One, two or three adult doses of platelets prepared from a single apheresis platelet donation. The donor's blood is separated into components with retention of the platelets and a portion of plasma. The red cells and other components of the blood are returned to the donor. Leucocyte depletion is performed during the collection. A platelet additive solution is added to achieve the pre-determined final platelet concentration. This component is irradiated prior to issue.
Indications	For treatment of bleeding due to either severely decreased platelet production or functionally abnormal platelets (e.g. inherited platelet disorders). Rarely may be used in treating patients with platelet consumption disorders but typically only for bleeding or selected invasive procedures (see contraindications). Platelets may be useful if given prophylactically to patients with rapidly falling or low platelet counts (usually < 10 x 10^9/L secondary to cancer or chemotherapy). May also be useful in selected cases of postoperative bleeding (e.g. platelet count < 50 x 10^9/L).
Contraindications	Should not be used if bleeding is unrelated to decreased numbers of platelets or abnormally functioning platelets. Do not use in patients with destruction of endogenous and exogenous platelets, such as in immune thrombocytopenic purpura (ITP), thrombotic thrombocytopenic purpura (TTP) or heparin-induced thrombocytopenia (HIT), unless the patient has a life-threatening haemorrhage.

Typical unit content

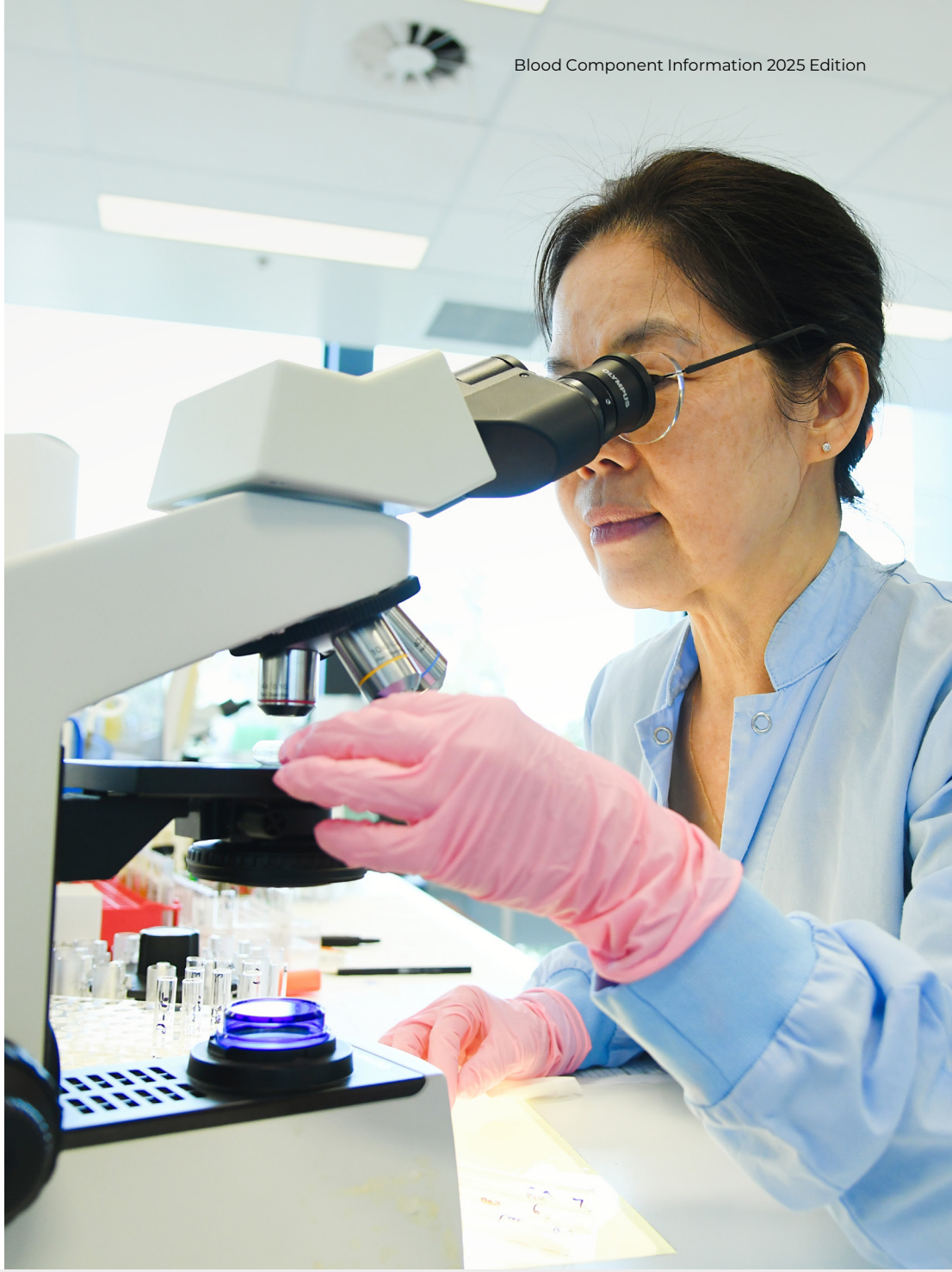
Data from:
1 January 2024 to
31 December 2024

Parameter	Mean (± 1 SD)	Specification
Volume (mL)	216 ± 9	100–400
Platelet count (x10^9/unit)	287 ± 35	> 200 to ≤ 450
pH (at expiry)	7.2 ± 0.1	6.4 – 7.4
Leucocyte count (x10^6/unit)	0.2 ± 0.2	< 1.0

Availability	Available in group O, A and B; and RhD positive and negative. Group AB must be requested in advance.
Shelf life	Seven days after collection.
Storage temperature	20°C to 24°C. Platelets components must be gently and continuously agitated in a single layer on a platelet agitator.
Dosage	One unit would be expected to increase the platelet count of a 70 kg adult by 20–40 x 10^9/L. The number of units prescribed depends on the patient's clinical situation. The usual dose in an adult is one unit. For prophylaxis, this dose may need to be repeated in one to three days because of the short life span of transfused platelets. Both immune and non-immune mechanisms may contribute to reduced platelet recovery and survival.
Administration	Transfuse through a blood administration set incorporating a standard (170–200 µm) filter. Transfusion of each unit may proceed as fast as tolerated but should be completed within four hours of removal from approved controlled storage. When RhD positive platelets are transfused to a RhD negative female of childbearing potential (including female children), prevention of RhD immunisation by use of RhD immunoglobulin should be considered. One 250 IU dose of RhD immunoglobulin, given intramuscularly, provides sufficient cover for six weeks of platelet transfusions.[12] If intravenous RhD immunoglobulin administration is required, Rhophylac® should be used.
Transfusion reactions	See Appendix IV: Transfusion reactions
Modifications	Irradiated, CMV seronegative, HLA-compatible, phenotyped, low anti-A/B. See Appendix II: Clinical indications for blood components which have been further manufactured or tested See Appendix III: Blood component modifications

Apheresis platelets
in platelet additive solution, leucocyte depleted *continued*

Comments	In addition to platelet-specific HPA (human platelet antigen) system antigens, platelets carry HLA class I antigens. Residual white blood cells which may be present in platelet units express both HLA class I and II antigens. Refractoriness to platelet transfusion may occur following HLA, or less commonly HPA, alloimmunisation. When transfused to a patient with an antibody specific to an expressed antigen, the survival time of the transfused platelets may be markedly shortened, and the patient may become either temporarily or permanently refractory to platelet transfusion. HLA-compatible or HPA-matched platelets may be indicated. Compatibility testing is not necessary in routine platelet transfusion. Platelet components should be ABO and RhD type compatible with the recipient. However, ABO-incompatible platelets may be used if ABO-compatible platelets are not available. In some patients (particularly children), residual plasma present in platelet units which are ABO-incompatible with the recipient's red cells may cause a positive direct antiglobulin test and possible low-grade haemolysis due to isoagglutinins present in the plasma. The priority for RhD negative platelets is RhD negative females of childbearing potential (including female children). Other RhD negative patients who require short-term platelet support should be given RhD negative platelets if available. Particular efforts should also be made to provide RhD negative platelets to young RhD negative male children. Resuspension of platelets in platelet additive solution rather than plasma reduces the risk associated with transfusion of large volumes of ABO-incompatible plasma and the incidence of adverse reactions to plasma proteins. Apheresis platelets that have low titre anti-A and/or anti-B pose a lower risk of haemolysis when transfusing ABO-incompatible platelets. In addition, group A platelets with the A_2 subgroup do not express significant amounts of A antigen and are therefore preferable to other group A platelets when transfusing group O and B recipients. Immunisation to donor red cell antigens may occur because of the presence of small but variable numbers of red cells in platelet units.



Paediatric apheresis platelets
in platelet additive solution, leucocyte depleted

Description	An adult dose apheresis platelet pack that is divided into three packs of equal volume for the purpose of reducing donor exposure in paediatric recipients and to minimise component wastage. This component is irradiated.
Indications	For treatment of bleeding due to either severely decreased platelet production or functionally abnormal platelets (e.g. inherited platelet disorders). Rarely may be used in treating patients with platelet consumption disorders but typically only for bleeding or selected invasive procedures (see contraindications). Platelets may be useful if given prophylactically to patients with rapidly falling or low platelet counts (usually < 10 x 10^9/L secondary to cancer or chemotherapy). May also be useful in selected cases of postoperative bleeding (e.g. platelet count < 50 x 10^9/L). Clinically indicated for small volume transfusions in infants and children.
Contraindications	Should not be used if bleeding is unrelated to decreased numbers of platelets or abnormally functioning platelets. Do not use in patients with destruction of endogenous and exogenous platelets, such as in immune thrombocytopenic purpura (ITP), thrombotic thrombocytopenic purpura (TTP) or heparin-induced thrombocytopenia (HIT), unless the patient has a life-threatening haemorrhage.

Typical unit content

Data from:
1 January 2024 to
31 December 2024

Parameter	Mean (± 1 SD)	Specification
Volume (mL)	55 ± 2	40–60
Platelet count (x10^9/unit)	71 ± 9	> 50
pH (at expiry)	7.0 ± 0.1	6.4 – 7.4
Leucocyte count (x10^6/unit)	NA	Initial unit prior to unit splitting < 1.0

Availability	Contact local Lifeblood centre regarding availability.
Shelf life	Seven days after collection.
Storage temperature	20°C to 24°C. Platelet components must be gently and continuously agitated in a single layer on a platelet agitator.
Dosage	One unit would be expected to increase the platelet count of an 18 kg child by 20 x 10^9/L. The number of units prescribed depends on the patient's clinical situation. For prophylaxis, this dose may need to be repeated in one to three days because of the short life span of transfused platelets. Both immune and non-immune mechanisms may contribute to reduced platelet recovery and survival.
Administration	Transfuse through a blood administration set incorporating a standard (170–200 µm) filter. Transfusion of each unit may proceed as fast as tolerated but should be completed within four hours of removal from approved controlled storage. When RhD positive platelets are transfused to a RhD negative female of childbearing potential (including female children), prevention of RhD immunisation by use of RhD Immunoglobulin should be considered. One 250 IU dose of RhD Immunoglobulin, given intramuscularly, provides sufficient cover for six weeks of platelet transfusions.[12] If intravenous RhD Immunoglobulin administration is required, Rhophylac® should be used.
Transfusion reactions	See Appendix IV: Transfusion reactions
Modifications	Irradiated, CMV seronegative, HLA-compatible, phenotyped, low anti-A/B. See Appendix II: Clinical indications for blood components which have been further manufactured or tested See Appendix III: Blood component modifications
Comments	See Apheresis platelets

Pooled platelets
in platelet additive solution, leucocyte depleted

Description	An adult dose of platelets obtained from a pool of buffy coats from ABO identical donors and resuspended in a platelet additive solution. The pool is then filtered to remove most leucocytes. This component is irradiated.
Indications	For treatment of bleeding due to either severely decreased platelet production or functionally abnormal platelets (e.g. inherited platelet disorders). Rarely may be used in treating patients with platelet consumption disorders but typically only for bleeding or selected invasive procedures (see contraindications). Platelets may be useful if given prophylactically to patients with rapidly falling or low platelet counts (usually < 10 x 10^9/L secondary to cancer or chemotherapy). May also be useful in selected cases of postoperative bleeding (e.g. platelet count < 50 x 10^9/L).
Contraindications	Should not be used if bleeding is unrelated to decreased numbers of platelets or abnormally functioning platelets. Do not use in patients with destruction of endogenous and exogenous platelets, such as in immune thrombocytopenic purpura (ITP), thrombotic thrombocytopenic purpura (TTP) or heparin-induced thrombocytopenia (HIT), unless the patient has a life-threatening haemorrhage.

Typical unit content

Data from:
1 January 2024 to
31 December 2024

Parameter	Mean (± 1 SD)	Specification
Volume (mL)	270 ± 12	> 160
Platelet count (x10^9/pool)	264 ± 37	> 200
pH (at expiry)	7.0 ± 0.1	6.4 – 7.4
Leucocyte count (x10^6/pool)	0.01 ± 0.2	< 1.0

Availability	Available in group O, A and B, and RhD positive and negative.
Shelf life	Seven days after collection.
Storage temperature	20°C to 24°C. Platelet components must be gently and continuously agitated in a single layer on a platelet agitator.
Dosage	One unit would be expected to increase the platelet count of a 70 kg adult by 20–40 x 10^9/L. The number of units prescribed depends on the patient's clinical situation. The usual dose in an adult is one pooled unit. For prophylaxis, this dose may need to be repeated in one to three days because of the short life span of transfused platelets. Both immune and non-immune mechanisms may contribute to reduced platelet recovery and survival.
Administration	Transfuse through blood administration set incorporating a standard (170–200 µm) filter. Transfusion of each unit may proceed as fast as tolerated but should be completed within four hours of removal from approved controlled storage. When RhD positive platelets are transfused to a RhD negative female of childbearing potential, prevention of RhD immunisation by use of RhD immunoglobulin should be considered. One 250 IU dose of RhD immunoglobulin, given intramuscularly, provides sufficient cover for six weeks of platelet transfusions.[12] If intravenous RhD immunoglobulin administration is required, Rhophylac® should be used.
Transfusion reactions	See Appendix IV: Transfusion reactions
Modifications	Irradiated, CMV seronegative. See Appendix II: Clinical indications for blood components which have been further manufactured or tested See Appendix III: Blood component modifications
Comments	See Apheresis platelets

Fresh frozen plasma (FFP)

Description	Clinical fresh frozen plasma (FFP) is either separated from a single unit of whole blood or collected by apheresis where the plasma is retained and the remaining elements are either returned to the donor or harvested for the appropriate component types. All apheresis-derived clinical FFP is split into two or three units of equal volume prior to freezing. Freezing of whole blood plasma must commence within 18 hours of collection and freezing of apheresis plasma must commence within six hours of collection. Contains all coagulation factors including approximately 200 IU of Factor VIII plus the other labile plasma coagulation factor, Factor V.
Indications	For patients with a coagulopathy who are bleeding or at risk of bleeding where a specific therapy such as vitamin K or factor concentrate is not appropriate or unavailable. May be indicated in bleeding patients who require replacement of labile plasma coagulation factors such as in massive transfusion, cardiac bypass, liver disease or acute disseminated intravascular coagulation (DIC). FFP may be indicated for patients with thrombotic thrombocytopenic purpura (TTP).
Contraindications	Do not use FFP when coagulopathy can be corrected more effectively with specific therapy, such as vitamin K or specific factor concentrates. Where Beriplex® AU 4 factor prothrombin complex concentrate (PCC) is used for rapid warfarin reversal in patients requiring emergency surgery or with life-threatening warfarin-related bleeding, FFP is not required. FFP may however be required for associated major haemorrhage as directed by local Major Haemorrhage Protocols.[13] Do not use FFP when blood volumes can be safely and adequately replaced with other volume expanders such as 0.9% sodium chloride solution, Hartmann's Solution, or appropriate colloids.

Typical unit content

Data from:
1 January 2024 to
31 December 2024

Whole blood derived **Parameter**	**Mean (± 1 SD)**	**Specification**
Volume (mL)	286 ± 12	250–310
FVIIIc (IU/mL)	1.08 ± 0.35	≥ 0.70
Apheresis derived **Parameter**	**Mean (± 1 SD)**	**Specification**
Volume (mL)	272 ± 4	250–310
FVIIIc (IU/mL)	1.32 ± 0.38	≥ 0.70

Availability	Available in all ABO groups.
Shelf life	12 months.
Storage temperature	–25°C or below. Once thawed, should be transfused immediately or stored at 2°C to 6°C for up to five days in accordance with ANZSBT guidelines.[2,14,15,16,17]
Dosage	The volume transfused depends on the clinical situation and patient size and should be guided by laboratory assays of coagulation function. Note: If using remaining supplies of Prothrombinex-VF® 3 factor prothrombin complex concentrate (PCC) for warfarin-related life-threatening bleeding (including critical organ and intracranial), refer to local guidelines regarding the dosing of FFP to deliver factor VII.
Administration	Thaw using an approved method. See Storage, transport, and handling. Mix thoroughly by inversion before use and transfuse through a blood administration set incorporating a standard (170–200 µm) filter. Transfusion of each unit may proceed as fast as tolerated but should be completed within four hours of removal from approved controlled storage.
Transfusion reactions	See Appendix IV: Transfusion reactions
Modifications	IgA deficient, secretor plasma Le(b+). See Appendix III: Blood component modifications
Comments	Compatibility tests before transfusion are not necessary. Plasma should be ABO-compatible with the recipient's red cells. Group O plasma should be restricted to group O recipients. Plasma components that have low titre anti-A or anti-B pose a lower risk of haemolysis when transfusing ABO incompatible components. Group AB plasma products, although suitable for patients of all ABO groups and typically used when the patient's group is unknown, are often in short supply and use may be restricted, for example to neonates. In emergencies or trauma situations when the patient's ABO group is unknown, group A plasma products with low titre anti-B may be used for adults as an alternative to group AB. Matching for RhD type is not necessary.

Paediatric fresh frozen plasma (FFP)

Description	Clinical fresh frozen plasma (FFP) from a single unit of whole blood is divided into four packs of equal volume for the purpose of reducing donor exposure in paediatric recipients and to minimise component wastage. Freezing of the plasma is commenced within 18 hours after collection of the whole blood. Contains all coagulation factors including the labile plasma coagulation factors VIII and V.
Indications	For patients with a coagulopathy who are bleeding or at risk of bleeding where a specific therapy such as vitamin K or factor concentrate is not appropriate or unavailable. May be indicated in bleeding patients who require replacement of labile plasma coagulation factors such as in massive transfusion, cardiac bypass, liver disease or acute disseminated intravascular coagulation (DIC). FFP may be indicated for patients with thrombotic thrombocytopenic purpura (TTP). Clinically indicated for small volume transfusions in infants and children.
Contraindications	Do not use FFP when coagulopathy can be corrected more effectively with specific therapy, such as vitamin K or specific factor concentrates. Where Beriplex® AU 4 factor prothrombin complex concentrate (PCC) is used for rapid warfarin reversal in patients requiring emergency surgery or with life-threatening warfarin-related bleeding, FFP is not required. FFP may however be required for associated major haemorrhage as directed by local Major Haemorrhage Protocols.[13] Do not use FFP when blood volumes can be safely and adequately replaced with other volume expanders such as 0.9% sodium chloride solution, Hartmann's Solution, or appropriate colloids.

Typical unit content Data from: 1 January 2024 to 31 December 2024	**Parameter**	**Mean (± 1 SD)**	**Specification**
	Volume (mL)	68 ± 3	60–80

Availability	Available in all ABO groups.
Shelf life	12 months.
Storage temperature	–25°C or below. Once thawed, should be transfused immediately or stored at 2°C to 6°C for up to five days in accordance with ANZSBT guidelines.[2,14,15,16,17]

Dosage	The volume transfused depends on the clinical situation and patient size, and should be guided by laboratory assays of coagulation function. Common paediatric dose is 10–15 mL/kg. Note: If using remaining supplies of Prothrombinex-VF® 3 factor prothrombin complex concentrate (PCC) for warfarin-related life-threatening bleeding (including critical organ and intracranial), refer to local guidelines regarding the dosing of FFP to deliver factor VII.
Administration	Thaw using an approved method. See Storage, transport, and handling. Mix thoroughly by inversion before use and transfuse through a blood administration set incorporating a standard (170–200 µm) filter. Transfusion of each unit may proceed as fast as tolerated but should be completed within four hours of removal from approved controlled storage.
Transfusion reactions	See Appendix IV: Transfusion reactions
Modifications	IgA deficient, secretor plasma Le(b+). See Appendix III: Blood component modifications
Comments	Compatibility tests before transfusion are not necessary. Plasma should be ABO compatible with the recipient's red cells. Group O plasma should be restricted to group O recipients. Plasma components that have low titre anti-A or anti-B pose a lower risk of haemolysis when transfusing ABO-incompatible components. Group AB plasma products, although suitable for patients of all ABO groups and typically used when the patient's group is unknown, are often in short supply and use may be restricted, for example to neonates. In emergencies or trauma situations when the patient's ABO group is unknown, group A plasma products with low titre anti-B may be used for adults as an alternative to group AB. Matching for RhD type is not necessary.

Cryodepleted plasma

Description	Cryodepleted plasma is the supernatant remaining after cryoprecipitate has been removed from whole blood derived FFP used for cryoprecipitate production. Contains most clotting factors in similar amounts to FFP but is deficient in Factor VIII, von Willebrand factor (VWF; the high molecular weight multimers are more thoroughly removed than the smaller multimers), fibrinogen, Factor XIII and fibronectin.
Indications	Cryodepleted plasma may be used for plasma exchange in thrombotic thrombocytopenic purpura (TTP). It may also be used as an alternative to FFP for the treatment of coagulopathy where there is no significant reduction in Factor VIII, fibrinogen, Factor XIII or VWF.
Contraindications	Do not use cryodepleted plasma when coagulopathy can be corrected more effectively with specific therapy, such as vitamin K or specific factor concentrates. Where Beriplex® AU 4 factor prothrombin complex concentrate (PCC) is used for rapid warfarin reversal in patients requiring emergency surgery or with life-threatening warfarin-related bleeding, cryodepleted plasma (as an alternative to FFP) is not required. Cryodepleted plasma may however be required for associated major haemorrhage as directed by local Major Haemorrhage Protocols.[13] Do not use cryodepleted plasma when blood volumes can be safely and adequately replaced with other volume expanders such as 0.9% sodium chloride solution, Hartmann's Solution, or appropriate colloids.

Typical unit content

Data from:
1 January 2024 to
31 December 2024

Parameter	Mean (± 1 SD)	Specification
Volume (mL)	246 ± 12	215–265

Availability	Available in all ABO groups.
Shelf life	12 months.
Storage temperature	–25°C or below. Once thawed, should be transfused immediately or stored at 2°C to 6°C for up to five days (unpublished Lifeblood data) in accordance with ANZSBT guidelines.[2,14,15,16,17]

Dosage	The volume transfused depends on the clinical situation and patient size, and should be guided by laboratory assays of coagulation function. Note: If using remaining supplies of Prothrombinex-VF® 3 factor prothrombin complex concentrate (PCC) for warfarin-related life-threatening bleeding (including critical organ and intracranial), refer to local guidelines regarding the dosing of cryodepleted plasma to deliver factor VII. One whole blood unit is equivalent to one unit of split apheresis cryodepleted plasma. A combination of whole blood and split apheresis units may also be used to form a dose, as both types are equivalent and interchangeable.
Administration	Thaw using an approved method. See Storage, transport, and handling. Mix thoroughly by inversion before use and transfuse through a blood administration set incorporating a standard (170–200 µm) filter. Transfusion of each unit may proceed as fast as tolerated but should be completed within four hours of removal from approved controlled storage.
Transfusion reactions	See Appendix IV: Transfusion reactions
Modifications	None.
Comments	Compatibility tests before transfusion are not necessary. Plasma should be ABO compatible with the recipient's red cells. Group O plasma should be restricted to group O recipients. Plasma components that have low titre anti-A or anti-B pose a lower risk of haemolysis when transfusing ABO-incompatible components. Group AB plasma products, although suitable for patients of all ABO groups and typically used when the patient's group is unknown, are often in short supply and use may be restricted, for example to neonates. In emergencies or trauma situations when the patient's ABO group is unknown, group A plasma products with low titre anti-B may be used for adults as an alternative to group AB. Matching for RhD type is not necessary.

Apheresis cryodepleted plasma

Description	Apheresis cryodepleted plasma is the supernatant remaining after cryoprecipitate has been removed from apheresis-derived FFP for cryoprecipitate production. Contains most clotting factors in similar amounts to FFP but is deficient in Factor VIII, von Willebrand factor (VWF; the high molecular weight multimers are more thoroughly removed than the smaller multimers), fibrinogen, Factor XIII and fibronectin. Large-dose apheresis cryodepleted plasma is not currently being manufactured. It has been replaced by split apheresis units, with three units produced from a single apheresis donation. One split apheresis unit is equivalent to one unit of whole blood cryodepleted plasma. Laboratories may have a mixed inventory of large-dose and split apheresis cryodepleted plasma units until stocks of large-dose units are used or expire.
Indications	Cryodepleted plasma may be used for plasma exchange in thrombotic thrombocytopenic purpura (TTP). It may also be used as an alternative to FFP for the treatment of coagulopathy where there is no significant reduction in Factor VIII, fibrinogen, Factor XIII or VWF.
Contraindications	Do not use cryodepleted plasma when coagulopathy can be corrected more effectively with specific therapy, such as vitamin K or specific factor concentrates. Where Beriplex® AU 4 factor prothrombin complex concentrate (PCC) is used for rapid warfarin reversal in patients requiring emergency surgery or with life-threatening warfarin-related bleeding, cryodepleted plasma (as an alternative to FFP) is not required. Cryodepleted plasma may however be required for associated major haemorrhage as directed by local Major Haemorrhage Protocols.[13] Do not use cryodepleted plasma when blood volumes can be safely and adequately replaced with other volume expanders such as 0.9% sodium chloride solution, Hartmann's Solution, or appropriate colloids.

Typical unit content

Data from:
1 January 2024 to
31 December 2024

*Final validation data

	Large dose (Manufacture discontinued)		Split apheresis (From January 2025)	
Parameter	**Mean (± 1 SD)**	**Specification**	**Mean (± 1 SD)***	**Specification**
Volume (mL)	757 ± 7	675–825	238 ± 4	215–265

Availability	Available in all ABO groups.
Shelf life	12 months.
Storage temperature	–25°C or below. Once thawed, should be transfused immediately or stored at 2°C to 6°C for up to five days (unpublished Lifeblood data) in accordance with ANZSBT guidelines.[2,14,15,16,17]
Dosage	The volume transfused depends on the clinical situation and patient size and should be guided by laboratory assays of coagulation function. Note: If using remaining supplies of Prothrombinex-VF® 3 factor prothrombin complex concentrate (PCC) for warfarin-related life-threatening bleeding (including critical organ and intracranial), refer to local guidelines regarding the dosing of cryodepleted plasma to deliver factor VII. A combination of split apheresis and whole blood units may also be used to form a dose, as both types are equivalent and interchangeable.
Administration	Thaw using an approved method. See Storage, transport, and handling. Mix thoroughly by inversion before use and transfuse through a blood administration set incorporating a standard (170–200 µm) filter. Transfusion of each unit may proceed as fast as tolerated but should be completed within four hours of removal from approved controlled storage.
Transfusion reactions	See Appendix IV: Transfusion reactions
Modifications	None.
Comments	Compatibility tests before transfusion are not necessary. Plasma should be ABO compatible with the recipient's red cells. Group O plasma should be restricted to group O recipients. Plasma components that have low titre anti-A or anti-B pose a lower risk of haemolysis when transfusing ABO-incompatible components. Group AB plasma products, although suitable for patients of all ABO groups and typically used when the patient's group is unknown, are often in short supply and use may be restricted, for example to neonates. In emergencies or trauma situations when the patient's ABO group is unknown, group A plasma products with low titre anti-B may be used for adults as an alternative to group AB. Matching for RhD type is not necessary.

Cryoprecipitate

Description	Cryoprecipitate is prepared by slow thawing of whole blood-derived FFP for cryoprecipitate production and recovering the cold-insoluble precipitate, which is then refrozen. Contains most of the original Factor VIII, von Willebrand factor (VWF), fibrinogen, Factor XIII and fibronectin. One whole blood unit is equivalent to one unit of split apheresis cryoprecipitate.		
Indications	Indicated for use in major haemorrhage protocols to replace fibrinogen and in the treatment of fibrinogen deficiency or dysfibrinogenaemia when there is clinical bleeding, an invasive procedure, trauma or disseminated intravascular coagulation (DIC).		
Contraindications	Should not be used to treat haemophilia, von Willebrand's disease or deficiencies of Factor XIII or fibronectin unless alternative therapies are unavailable.		
Typical unit content Data from: 1 January 2024 to 31 December 2024	**Parameter**	**Mean (± 1 SD)**	**Specification**
	Volume (mL)	37 ± 2	30–40
	Fibrinogen (mg/unit)	399 ± 123	≥ 140
	FVIIIc (IU/unit)	143 ± 25	≥ 70
	VWF (IU/unit)	247 ± 42	> 100
Availability	Available in all ABO groups.		
Shelf life	12 months.		
Storage temperature	–25°C or below. Thawed cryoprecipitate should be maintained at 20°C to 24°C until transfused.[14,18] Once thawed, should be used within six hours if it is a closed single unit, or within four hours if it is an open system or units have been pooled.[14,18]		

Dosage	Typically, one unit per 5–10 kg body weight would be expected to increase the fibrinogen concentration by 0.5–1.0 g/L. The volume transfused depends on the clinical situation and patient size and should be guided by laboratory assays of coagulation function. A typical adult dose is nine units of whole blood-derived cryoprecipitate, approximately equivalent to a fibrinogen dose of 3–4g. See Appendix 1: Patient blood management guidelines (Patient blood management guideline for adults with critical bleeding and Module 6: Neonatal and paediatrics).[1] A combination of whole blood and split apheresis cryoprecipitate units may also be used to form a dose, as both types are equivalent and interchangeable. In the steady state, the half-life of fibrinogen is three to five days. Dosing schedules of cryoprecipitate infusions every three days may be appropriate for patients with congenital hypofibrinogenemia.
Administration	Thaw using an approved method. See Storage, transport, and handling. If choosing to pool multiple units, the precipitate in each concentrate should be mixed well with 10–15 mL of diluent to ensure complete removal of all material from the container. The preferred diluent is 0.9% sodium chloride Injection (USP). Mix thoroughly by inversion before use and transfuse through a blood administration set incorporating a standard (170–200 µm) filter. Transfusion of each unit may proceed as fast as tolerated but should be completed within four hours of removal from approved controlled storage.
Transfusion reactions	See Appendix IV: Transfusion reactions
Modifications	None.
Comments	When reviewing international studies of fibrinogen replacement, ensure that the current fibrinogen content in Australian cryoprecipitate is considered when estimating equivalence. Due to differences in manufacturing process between Blood Services internationally, fibrinogen content per 'unit' (pack) and per mL can vary considerably. Compatibility tests before transfusion are not necessary. Preferably ABO compatible with the recipient's red cells; however, ABO-incompatible cryoprecipitate can be used with caution, particularly with large volumes. If a large volume of ABO-incompatible cryoprecipitate is used, the recipient may develop a positive direct antiglobulin test and, very rarely, mild haemolysis. Plasma components that have low titre anti-A or anti-B pose a lower risk of haemolysis. Matching for RhD type is not necessary.

Apheresis cryoprecipitate

Description

Apheresis cryoprecipitate is prepared by slow thawing of apheresis-derived FFP for cryoprecipitate production and recovering the cold-insoluble precipitate, which is then refrozen.

Contains most of the original Factor VIII, von Willebrand factor (VWF), fibrinogen, Factor XIII and fibronectin.

Large-dose apheresis cryoprecipitate is not currently being manufactured. It has been replaced with split apheresis units, with three units produced from a single donation. One split apheresis unit is equivalent to one unit of whole blood cryoprecipitate.

Laboratories may have a mixed inventory of large-dose and split apheresis cryoprecipitate units until stocks of large-dose units are used or expire.

Indications

Indicated in the treatment of fibrinogen deficiency or dysfibrinogenaemia when there is clinical bleeding, an invasive procedure, trauma or disseminated intravascular coagulation (DIC).

Contraindications

Should not be used to treat haemophilia, von Willebrand's disease or deficiencies of Factor XIII or fibronectin, unless alternative therapies are unavailable.

Typical unit content

Data from:
1 January 2024 to
31 December 2024

*Final validation data

	Large dose (Manufacture discontinued)		Split apheresis (From January 2025)	
Parameter	**Mean (± 1 SD)**	**Specification**	**Mean (± 1 SD)***	**Specification**
Volume (mL)	60 ± 2	54–66	33 ± 3	30–40
Fibrinogen (mg/unit)	1173 ± 334	≥ 140	395 ± 116	≥ 140
FVIIIc (IU/unit)	372 ± 78	≥ 70	178 ± 22	≥ 70
VWF (IU/unit)	620 ± 102	> 100	261 ± 52	> 100

Availability

Available in all ABO groups.

Shelf life

12 months.

Storage temperature

–25°C or below.

Thawed cryoprecipitate should be maintained at 20°C to 24°C until transfused.[14,18] Once thawed, should be used within six hours if it is a closed single unit, or within four hours if it is an open system or units have been pooled.[14,18]

Dosage

Typically, one unit per 10–20 kg body weight would be expected to increase the fibrinogen concentration by 0.5–1.0 g/L.

The volume transfused depends on the clinical situation and patient size, and should be guided by laboratory assays of coagulation function.

A typical adult dose is nine split units of apheresis-derived cryoprecipitate (or three large-dose apheresis units where still available), approximately equivalent to a fibrinogen dose of 3–4g. See Appendix 1: Patient blood management guidelines (Patient blood management guideline for adults with critical bleeding and Module 6: Neonatal and paediatrics).[1]

A combination of split apheresis and whole blood units may also be used to form a dose, as both types are equivalent and interchangeable.

In the steady state, the half-life of fibrinogen is three to five days. Dosing schedules of cryoprecipitate infusions every three days may be appropriate for patients with congenital hypofibrinogenemia.

Administration

Thaw using an approved method. See Storage, transport, and handling.

If choosing to pool multiple units, the precipitate in each concentrate should be well mixed with 10–15 mL of diluent to ensure complete removal of all material from the container. The preferred diluent is 0.9% sodium chloride Injection (USP).

Mix thoroughly by inversion before use and transfuse through a blood administration set incorporating a standard (170–200 µm) filter.

Transfusion of each unit may proceed as fast as tolerated but should be completed within four hours of removal from approved controlled storage.

Transfusion reactions

See Appendix IV: Transfusion reactions

Modifications

None.

Comments

When reviewing international studies of fibrinogen replacement, ensure that the current fibrinogen content in Australian cryoprecipitate is considered when estimating equivalence. Due to differences in manufacturing process between Blood Services internationally, fibrinogen content per 'unit' (pack) and per mL can vary considerably. Compatibility tests before transfusion are not necessary. Preferably ABO compatible with the recipient's red cells; however, ABO-incompatible cryoprecipitate can be used with caution, particularly with large volumes.

If a large volume of ABO-incompatible cryoprecipitate is used, the recipient may develop a positive direct antiglobulin test and, very rarely, mild haemolysis. Matching for RhD type is not necessary.

Appendices



Appendix I: Patient Blood Management Guidelines

The National Blood Authority's series of six Patient Blood Management (PBM) Guidelines support a comprehensive evidence-based national patient blood management strategy. The NBA has commenced a review of the guidelines, but the original modules remain available to guide practice.[1]

PBM principles

Patient blood management (PBM) improves patient outcomes by ensuring that the focus of the patient's medical and surgical management is on improving and conserving the patient's own blood.

When a PBM approach is used, patients usually require fewer transfusions, reducing the risk of transfusion-associated complications.

If blood components are likely to be indicated, transfusion should not be a default decision. Instead, the decision on whether to transfuse should carefully consider the full range of available therapies and balance the evidence for efficacy and improved clinical outcome against the potential risks.

PBM incorporates proactive treatment regimens which are tailored to suit individual patients, using a multidisciplinary team approach to conserve a patient's own blood. Consideration should be given to the use of pharmaceutical agents and medical devices to reduce the need for allogeneic blood transfusion.

Patient blood management guideline for adults with critical bleeding

To assist and guide healthcare professionals in making clinical decisions when managing patients with critical bleeding who require, or are likely to require, massive transfusion.

View the module here →

Module 2
Perioperative

To inform healthcare practitioners, health educators, and health service managers and policy-makers about the pre, intra and postoperative care of patients undergoing surgery or invasive procedures, particularly those in which blood loss is anticipated.

View the module here →

Module 3
Medical

To assist and guide clinical decisions and coordination of healthcare across the primary, secondary and tertiary care settings for patients with acute or chronic medical conditions requiring haematological intervention.

View the module here →

Module 4
Critical Care

To assist and guide healthcare professionals in making clinical decisions when managing patients requiring critical care.

View the module here →

Module 5
Obstetrics and Maternity

To assist and guide healthcare professionals in making clinical decisions when managing pregnant and postpartum women.

View the module here →

Module 6
Neonatal and Paediatrics

To assist and guide healthcare professionals in making clinical decisions about blood management in neonatal and paediatric patients.

View the module here →

Appendix II: Clinical indications for blood components that have been further manufactured or tested

CMV seronegative[19, 20, 21, 22]

Description	Transmission of CMV disease is associated with cellular blood components. CMV seronegative red cells and platelets are identified by testing selected donations for CMV antibodies, using a CMV test approved for donor screening. FFP, cryoprecipitate and other plasma-derived blood components do not transmit CMV, therefore CMV testing is not required for these components.
Indications	CMV seronegative recipients at risk of severe CMV disease.
Patient groups	CMV seronegative products should be used for the following clinical indications: pregnant women regardless of CMV status who require regular elective transfusions during pregnancy (but not during delivery); recipients of intrauterine transfusion (IUT); neonates (up to 28 days post expected date of delivery); granulocyte transfusions for CMV negative patients. For other groups: solid organ transplants, haemopoietic stem cell transplants (HSCT), haematology /oncology patients and immunodeficient patients, including those with HIV, leucodepleted blood products are considered CMV safe but the decision to use will be dictated by local clinical policies.
Comments	Where CMV seronegative blood components are indicated, select CMV seronegative components whenever possible. If not available, leucodepleted components are considered to offer a high level of safety in preventing CMV transmission. Whether using a combination of leucocyte depletion and CMV seronegative components provides any additional safety benefits is unknown.

Frozen red cells

Description	Glycerol is added to red cells as a cryoprotectant before freezing at between –65°C and –80°C. Frozen red cells may be stored for up to 30 years, although longer periods may be possible if there is a particular need for specific units.
Indications	Treating anaemia or blood loss.
Patient groups	Patients with rare red cell phenotypes, or multiple red cell antibodies and for autologous collections when liquid-preserved blood cannot fulfil demands.
Comments	Prior to transfusion, glycerol is removed from the thawed component by washing the cells with sodium chloride. After washing, the red cells are resuspended in SAG-M additive solution and must be used within 24 hours. There will be some loss of red cells during the freezing and thawing process. When requesting frozen red cells it should be noted that thawing and processing time is several hours.

Appendix II: Clinical indications for blood components that have been further manufactured or tested continued

Irradiation[23]

Description	Irradiation is used to inactivate viable T-lymphocytes found in red cells, platelets and granulocytes which are the immediate cause of transfusion-associated graft versus host disease (TA-GVHD), a rare, but almost universally fatal complication of transfusion. The minimum irradiating dose should be 25 Gy, with no part of the component receiving greater than 50 Gy. A radiosensitive indicator label attached to the pack indicates irradiation was satisfactorily performed.
Indications	Prevention of TA-GVHD in susceptible recipients.
Patient groups	Intrauterine transfusion; neonatal exchange transfusions; neonatal small volume ("top- up") transfusions if infant has received prior IUT; congenital and acquired immunodeficiencies in infants and children; remission induction and consolidation therapy for acute leukaemia and chemotherapy of similar intensity for other malignancies; allogeneic stem cell transplantation; autologous stem cell transplant; haematopoietic stem cell donors (including autologous and T cell donors); chimeric antigen receptor T (CAR-T) cell recipients; Hodgkin Lymphoma; aplastic anaemia; patients receiving purine analogues or bendamustine for malignant or non-malignant disorders; patients receiving alemtuzumab for haematological neoplasms; HLA-matched (HLA-compatible) donors; related donors; patients receiving transfusion due to radiation injury.
Comments	Irradiation of red cells increases the rate of efflux of extracellular potassium. In considering the clinical significance of this, both the speed and volume of the transfusion, as well as the age of the blood, must be considered. Red cells (and washed red cells) may be irradiated at any time up to 14 days after collection, and post irradiation shelf life is 14 days. Red cells for IUT and exchange transfusion must be less than five days old at irradiation and once irradiated must be used within 24 hours. Irradiated paediatric red cells must be less than 14 days at irradiation and once irradiated must be used within 48 hours. Platelets can be irradiated at any stage during their seven-day shelf life without any change to their original seven-day expiry date. Granulocytes for all recipients should be irradiated as soon as possible after production and transfused with minimal delay. Stem cells, donor T lymphocytes, CAR-T cells or other cellular products required to engraft, whether allogeneic or autologous, must not be irradiated as they will be rendered ineffective. Red cells cold stored for 21 days or more are irradiation equivalent. Red cells stored for 14 days or more may be considered TA-GVHD safe and utilised for patients at risk of TA-GVHD when irradiation equivalent red cells are not available. In patients at risk of TA-GVHD who need emergency transfusion, the use of the shortest expiry (oldest) suitable red cells is acceptable if irradiated (or equivalent) units are not available.

Phenotyped and/or genotyped red cell and platelet components

Description	For patients requiring specific antigen-negative or matched red cell or platelet components due to, or to prevent, alloimmunisation.
Indications	Prevention or management of alloimmunisation to red cell or platelet (HPA or HLA) antigens.
Patient groups	Patients with red cell or platelet alloantibodies, patients receiving long-term transfusion support, patients with warm autoimmune haemolytic anaemia (AIHA), patients receiving therapeutic monoclonal antibodies, for example anti-CD38 or anti-CD47. HLA compatible red cells may be provided for non-urgent transfusions in selected patients awaiting living donor kidney transplantation.
Comments	None.

Red cells for intrauterine transfusion (IUT)

Description	A hyper-concentrated red cell component less than five days old with a haematocrit of 0.70–0.85 obtained by removing most of the plasma/additive solution. The red cells may be resuspended in additive solution to achieve the desired haematocrit. The final volume is printed on the component label. Red cells for IUT must be irradiated.[23]
Indications	Treatment of fetal anaemia primarily associated with haemolytic disease of the fetus and newborn (HDFN).
Patient groups	Fetuses with anaemia.
Comments	ABO, RhD compatible with both mother and fetus, K negative. Should be antigen-negative for maternal alloantibodies, IAT crossmatch compatible with the maternal plasma and CMV seronegative. If the fetal blood group is unknown use group O, RhD negative red cells. Once irradiated the component must be used within 24 hours.

Appendix III: Blood component modifications

Modifier text	Explanation
Autol release – see disclaimer	This autologous component has tested positive for one or more viral markers but has been released upon request by the patient's physician. A disclaimer form will accompany these components.
CMV negative	The originating donor sample/donation has been tested for CMV antibody and is negative.
For intrauterine transfusion	A hyper-concentrated red cell component with a haematocrit of 0.70–0.85. After preparation the expiry is reduced to 48 hours and the component must be irradiated before use. Once irradiated the component must be used within 24 hours.
IgA deficient	The originating donor sample/donation, or the donor previously, has been tested and is IgA deficient.
Low Anti-A/B	The originating donor sample/donation has low levels of anti-A/B haemolysins.
Not for neonatal use	The component has been deemed unsuitable for neonatal use due to the presence of red cell antibodies (low titre only). It should not be transfused to a neonate.
Not NAT tested	Due to extenuating circumstances, (e.g. machine failure or specific clinical demand) this component has been released without nucleic acid technology (NAT) testing being performed. A disclaimer form will accompany these components.
Phenotype reserve	The originating donor sample/donation or, the donor previously, has had an extended phenotype performed and forms part of a panel of cells reserved for patients with antibodies or where antigen negative blood is otherwise specifically required.
Secretor plasma Le(b+)	The component is from an Le(a-b+) donor and contains soluble H and Lewis antigens (and A or B antigens if the donor is group A or B). May be used to neutralise a recipient's Lewis or A or B antibodies. Suitable for transfusion.
Suitable for research	The component is deemed unsuitable for clinical use, but may be used for research purposes. (These components will never be issued for transfusion).



Appendix IV: Transfusion reactions

Adverse reactions associated with transfusion of blood and blood components may be broadly classified as acute or delayed, and immunological or non-immunological.

Transfusion reactions

- Acute haemolytic transfusion reaction
- Allergic reaction
- Anaphylaxis or anaphylactoid reaction
- Delayed haemolytic transfusion reactions (DHTR)
- Febrile non-haemolytic transfusion reaction (FNHTR)
- Iron overload
- Massive transfusion complications
- Post-transfusion purpura (PTP)
- Transfusion-associated circulatory overload (TACO)
- Transfusion-associated graft-versus-host disease (TA-GVHD)
- Transfusion-related acute lung injury (TRALI)
- Transfusion-related immune modulation (TRIM)
- Transfusion-transmitted bacterial infection (TTBI)
- Transfusion-transmitted malaria
- Transfusion-transmitted viral infection
- Transfusion-transmitted infections: Other (TTI)
- Variant Creutzfeldt-Jakob Disease (vCJD)

Resources

Further information on aetiology, incidence, diagnosis, management, and prevention of transfusion reactions can be found in a variety of sources, including:

- Lifeblood's website
- Lifeblood's iTransfuse App
- Kopko PM, ed. Transfusion Reactions[24]
- Serious Hazards of Transfusion (SHOT)
- AABB Technical Manual[25,26]
- World Health Organisation Educational Modules on Clinical Use of Blood[8], and
- Australian and New Zealand Society of Blood Transfusion and Royal College of Nursing of Australia Guidelines for the Administration of Blood Products.[3]

Residual risk estimates for transfusion-transmitted infections

Current residual risk estimates for transfusion-transmitted infection in Australia are provided in Appendix V and at lifeblood.com.au



Appendix IV: Transfusion reactions continued

Acute haemolytic transfusion reaction (AHTR)

Usual aetiology	Immunologic incompatibility between transfused red cells and a recipient red cell antibody causing haemolysis or accelerated clearance of transfused red cells. Most severe reactions are associated with transfusion of ABO-incompatible blood due to clerical errors, patient identification errors such as improper sample labelling, testing errors or administering blood to the wrong patient. Occasionally transfusion of platelets or plasma components containing high titre anti-A and/or anti-B antibodies may cause clinically significant haemolysis if the plasma is incompatible with the recipient's red cells. May also be due to non-immune haemolysis, e.g. physical or chemical damage to transfused red cells, effects of drugs or hypotonic solutions co-administered with transfusion, effects of bacterial toxins, thermal injury due to freezing or overheating etc.
Incidence	1:40,000 for ABO/RhD incompatible transfusion, 1:76,000 for acute haemolytic reactions, 1:1,800,000 for fatal haemolytic reactions.[26]
Main clinical features	Most commonly fever, chills and haemoglobinuria; symptoms may also include tachycardia, dyspnoea, chest, back or flank pain, renal failure, abnormal bleeding or shock. Instability of blood pressure is frequent. In anaesthetised patients, hypotension and evidence of disseminated intravascular coagulation (DIC) may be the first sign.
Investigation	Clinical assessment to exclude other clinical or physical causes of haemolysis or shortened red cell survival. Clerical check of ABO typing of patient and unit. Blood group, antibody screen, crossmatch and direct antiglobulin test (DAT) on pre and post-transfusion specimens, renal function, tests for haemolysis (e.g. lactate dehydrogenase [LDH], haptoglobin, bilirubin), coagulation profile including fibrinogen etc.
Intervention	Stop transfusion immediately. Seek urgent assistance. Maintain blood pressure and renal output. Inform the Transfusion Service Provider.

Allergic reaction

Usual aetiology	Unclear; hypersensitivity to allergens or plasma proteins, rarely to donor medication.
Incidence	1–3% of transfusions of plasma-containing components.[24,26]
Main clinical features	Urticarial lesions, but may also include pruritus, erythema, flushing, or mild upper respiratory symptoms (cough, wheezing), nausea, vomiting, abdominal cramps or diarrhoea.
Investigation	Usually none.
Intervention	Stop transfusion. Give antihistamine. Continue transfusion at a slower rate when the reaction abates. Consider premedication for recurrent reactions. Washed red cells may be considered for severe reactions.

Appendix IV: Transfusion reactions continued

Anaphylaxis or anaphylactoid reaction

Usual aetiology	The majority of these reactions have been reported in IgA-deficient patients who have anti-IgA (of IgE class). May be due to IgE-mediated allergy to other plasma proteins, rarely to donor medication, etc.
Incidence	1:20,000–50,000 transfusions.[24,26]
Main clinical features	Often rapid onset. Reactions usually begin within 1–45 minutes after the start of the transfusion. Symptoms include severe hypotension, hypoxia, coughing, bronchospasm, laryngospasm, angioedema, urticaria, nausea, abdominal cramps, vomiting, diarrhoea, shock and loss of consciousness. May be fatal.
Investigation	Test pretransfusion sample for IgA and IgA antibodies. Consider tryptase testing.
Intervention	Stop transfusion immediately. Maintain airway and IV line, support blood pressure. Administer oxygen and adrenaline. The benefit of corticosteroids in anaphylaxis is unproven. Consult a haematologist before administering additional blood products. Use IgA deficient or washed components if patient is confirmed IgA deficient and future transfusion is required.

Delayed haemolytic transfusion reactions (DHTR)

Usual aetiology	Usually occur in patients previously alloimmunised to red cell antigens by transfusion or pregnancy. The red cell alloantibody is typically too weak to be detectable in pretransfusion testing. Transfusion of antigen-positive red cells provokes an anamnestic response and a rapid rise in antibody titre. Usual timeframe is 2 to 14 days after transfusion.
Incidence	Varying estimates 1:500 to 1:10,000.[24,26]
Main clinical features	Signs may include unexplained fever, unexplained decrease in haemoglobin or poor haemoglobin increment post-transfusion, jaundice and development of a positive DAT.
Investigation	DAT and red cell antibody screen. Liver function tests. Markers of haemolysis (reticulocyte count, bilirubin, LDH, haptoglobin, urinary haemosiderin, etc.).
Intervention	Most delayed haemolytic reactions have a benign course and require no treatment. Provide antigen negative blood if further transfusion is needed.

Appendix IV: Transfusion reactions continued

Febrile non-haemolytic transfusion reaction (FNHTR)

Usual aetiology	Accumulation of soluble factors (e.g. cytokines) during storage of cellular components; or recipient antibodies to white blood cell or other antigens.
Incidence	0.1–1% of transfusions with universal leucocyte depletion.[26] Most frequently in patients alloimmunised by transfusion or pregnancy.
Main clinical features	Pyrexia (temperature of ≥ 38°C and rise of ≥ 1°C from baseline) shortly after transfusion in the absence of any other pyrexic stimulus. May be associated with chills, rigors, nausea, vomiting, and headache.
Investigation	FNHTR is a diagnosis of exclusion. Similar symptoms may occur in other serious transfusion reactions e.g. acute haemolytic reaction, bacterial contamination and TRALI.
Intervention	Stop transfusion. Consider and exclude other causes. Give antipyretic. Recommencement of the transfusion, at a slow rate, is possible if other causes of a fever have been excluded.

Iron overload

Usual aetiology	Long-term complication of repeated red cell transfusions with iron deposition in organs. Each transfusion contributes about 250 mg of iron.
Incidence	Increased morbidity and mortality associated with cumulative doses of 20 or more units.[26]
Main clinical features	Symptoms and signs of organ damage or failure, especially hepatic and cardiac, and arthropathy.
Investigation	Patient history of chronic red cell transfusions. Serum ferritin and transferrin saturation. Organ imaging and/or liver biopsy.
Intervention	Iron chelating agents for treatment and for prevention where anticipated.

Massive transfusion complications

Usual aetiology	Rapid infusion of large volumes of cold blood can depress the body temperature and result in metabolic complications such as acidosis, hyperkalaemia, hypocalcaemia, etc. Rarely seen outside the massive transfusion setting. Trauma-induced haemostatic changes may further exacerbate complications of massive transfusion.
Incidence	Variable.
Main clinical features	Depends on clinical situation. Complications include metabolic and haemostatic abnormalities.
Investigation	Blood gas analysis (including acid-base balance, pH, electrolytes, ionised calcium), monitoring of coagulation profiles, possible thromboelastography and cardiac monitoring.
Intervention	Massive transfusion should follow the national Patient blood management guidelines for adults with critical bleeding and institutional protocols. Use blood warmers where appropriate. Consider calcium replacement and tranexamic acid infusions.

Post-transfusion purpura (PTP)

Usual aetiology	Alloimmunisation to platelet-specific antigens, most often human platelet antigen 1a (HPA-1a); however antibodies to other HPA antigens have also been implicated. While the immune specificity may be to a platelet-specific antigen the patient lacks, both autologous and allogeneic platelets are destroyed.
Incidence	Rare.
Main clinical features	Development of dramatic, sudden and self-limiting thrombocytopenia sufficient to cause purpura, petechiae, and clinically significant bleeding. Occurs typically 7 to 10 days after a blood transfusion, usually in a patient with a history of sensitisation by either pregnancy or transfusion.
Investigation	Demonstrate antiplatelet antibody.
Intervention	Intravenous immunoglobulin. Antigen-negative platelets may be indicated if platelet transfusion is required but this is controversial.

Appendix IV: Transfusion reactions continued

Transfusion-associated circulatory overload (TACO)

Usual aetiology	Volume overload due to rapid or large volume transfusion in patients with diminished cardiac reserve or chronic anaemia (usually children or elderly patients).
Incidence	Approximately 1% – 2.7% of patients receiving transfusions.[24, 26]
Main clinical features	Includes dyspnoea, orthopnoea, cyanosis, tachycardia, increased venous pressure and pulmonary oedema. May develop during or within 12 hours of transfusion.
Investigation	Clinical assessment, including a chest x-ray to check for presence of pulmonary oedema, although similar findings can be seen in TRALI. Raised levels of brain natriuretic peptide (BNP) may be informative.
Intervention	Stop transfusion. Administer oxygen, diuretics and keep patient upright. In susceptible patients, further transfusions should be administered slowly.

Transfusion-associated graft-versus-host disease (TA-GVHD)

Usual aetiology	Viable T lymphocytes in the transfused component proliferate in the recipient, engraft and mount a destructive immune response against the recipient's tissues. Pathogenesis is unclear. Immunocompromised recipients may be at increased risk but also seen in immunocompetent recipients, typically individuals heterozygous for an HLA haplotype for which the donor is homozygous, for example, genetically homogeneous populations such as the Japanese, recipients of HLA-matched components or blood from a close relative. However, in many cases particularly in immunocompetent individuals, there may be no identifiable differences between donor and recipient HLA types.
Incidence	Rare.
Main clinical features	Develops 4 to 30 days after transfusion. Characteristically fever, rash, gastrointestinal symptoms, liver injury and pancytopenia. Leads to profound marrow aplasia. Usually fatal.
Investigation	Biopsy of involved tissue (skin, liver, gut and possibly bone marrow), HLA typing (patient and donor) and chimerism studies. Demonstrate engraftment or chimerism of donor lymphocytes.
Intervention	Supportive care. Supportive or immunosuppressive therapies. Haemopoietic stem cell transplantation is rarely a viable option. However there is currently no effective treatment and is nearly always fatal. For at-risk patients, ensure irradiation of any cellular blood components.[23]

Appendix IV: Transfusion reactions continued

Transfusion-related acute lung injury (TRALI)

Usual aetiology	The most widely held pathogenesis theory is that human leucocyte antigen (HLA) or human neutrophil antigen (HNA) antibodies found in the donor's plasma are directed against the recipient's neutrophils. The antigen-antibody reaction activates neutrophils in the lung microcirculation, releasing oxidases and proteases that damage blood vessels resulting in leakage. Biological response modifiers, such as biologically active lipids can accumulate in some cellular components during storage and may also induce TRALI in susceptible patients.
Incidence	Variously reported. 1:1,200–190,000 transfusions.[26] TRALI is thought to be the most common cause of transfusion-associated fatalities.
Main clinical features	Onset of fever, tachycardia, hypotension, hypoxia ($PaO_2/FiO_2 \leq 300$ or $SpO_2 < 90\%$ on room air) and pulmonary oedema leading to respiratory failure during or within six hours of transfusion.
Investigation	Clinical assessment and investigation (e.g. chest x-ray, oxygen saturation and laboratory investigations). TRALI is a clinical diagnosis. Diagnosis may be supported by retrospectively demonstrating HLA or HNA antibodies in the donor together with a positive crossmatch.
Intervention	Stop transfusion immediately. Provide cardiovascular and respiratory support. Inform Lifeblood.

Transfusion-related immune modulation (TRIM)

Usual aetiology	A transient immunosuppression in recipients which may occur following transfusion of allogeneic blood, however the exact mechanism is yet to be elucidated. Possibly mediated by donor white cells releasing cytokines during storage, which leads to immune modulation.
Incidence	Not known.
Main clinical features	No specific signs or symptoms. Suggested increased risk of post-operative bacterial infection and tumour recurrence in cancer patients.
Investigation	No specific investigation process has been defined.
Intervention	Leucodepletion may have possible benefits.

Appendix IV: Transfusion reactions continued

Transfusion-transmitted bacterial infection (TTBI)

Usual aetiology	Bacteria may enter the blood during collection or preparation of components. Occasionally bacteria may enter due to contamination of ports whilst thawing frozen components in a water bath. Both gram-positive and gram-negative organisms have been implicated with serious morbidity and mortality occurring most frequently with gram-negative bacteria. Greatest risk is for platelets stored at room temperature. Organisms capable of multiplying at low temperatures and those using iron as a nutrient are most often associated with red cell contamination, especially *Yersinia enterocolitica*. To minimise the risk of bacterial contamination of platelets, Lifeblood screens all platelet components using an automated microbial detection system.
Incidence	For clinically apparent reactions, variously reported in the international literature to be 1:10,000 – 1:100,000 for platelets and at least 1:500,000 for red cells.[24,25,27,28] In comparison, Australian data show clinically apparent reactions occur at a rate of approximately 1:250,000 for platelets and approximately 1:2.5 million for red cells.[29]
Main clinical features	Onset of high fever, severe chills, rigors, hypotension, tachycardia, dyspnoea or circulatory collapse during or soon after transfusion should suggest the possibility of bacterial contamination and/or endotoxin reaction. Can be acute, severe and life-threatening. In severe cases, patient may develop shock with accompanying renal failure and DIC. This reaction may be fatal.
Investigation	Clinical assessment. Blood cultures from the patient. Culture and Gram stain of blood component. Keep blood bag and giving set (sealed) for further investigation.
Intervention	Stop transfusion immediately. Seek urgent medical assistance as this may become an emergency. Provide cardiovascular support. Start broad spectrum antibiotics once cultures have been taken. Send blood pack to Transfusion Service Provider. Inform Lifeblood to ensure quarantining and testing of related components from the same donation/donor.

Transfusion-transmitted malaria

Usual aetiology	Malaria is transmitted to humans through the bite of a mosquito infected with a parasite species of the *Plasmodium* genus or through blood transfusion from a donor infected with malaria. Although rare, transfusion-transmitted malaria continues to pose a risk. In order to minimise this risk, all potential blood donors are subjected to stringent screening procedures, including collection of a comprehensive medical and travel history as part of the donor assessment process. Lifeblood also performs malarial antibody screening on donors with a potential malarial exposure risk.
Incidence	Estimated residual risk per unit for malaria is less than 1 in 1 million units transfused.[30]
Main clinical features	Patients can present with fever, headache, nausea, vomiting and hypotension.
Investigation	Clinical assessment. Laboratory investigation including preparation of thick and thin blood films.
Intervention	Treat specific diagnosis. Treat specific parasite with antimalarial drugs. Inform Lifeblood.

Appendix IV: Transfusion reactions continued

Transfusion-transmitted viral infection

Usual aetiology	Transfusion-transmitted viral infections, such as hepatitis, HIV and HTLV may occur through donations made by donors in the window period of their infection. CMV is transfusion transmissible mainly through latent viruses located in white blood cells found in cellular blood components. There are also some viruses for which there are no routine tests available in Australia, such as Dengue Virus (Dengue Fever) and Parvovirus B19 (B19 viral infection). In addition, Lifeblood does not test donors for some viruses. For example, Chikungunya virus is such a low risk, that travel deferrals are an effective precaution.
Incidence	The risk per unit for HIV, hepatitis B, hepatitis C and HTLV I/II are all less than 1:1,000,000. For more information refer to Appendix V.
Main clinical features	Variable severity from asymptomatic to fatal. Features of specific clinical infection.
Investigation	Clinical assessment. Liver function tests. Specific testing for viral markers.
Intervention	Treatment for specific diagnosis, if available. The risk of CMV transmission may be reduced by requesting CMV seronegative blood components for specific at risk patient groups or using leucodepleted (CMV safe) components when clinically acceptable. Inform Lifeblood.

Transfusion-transmitted infections - Other (TTI)

Usual aetiology	Lifeblood's mandatory testing includes the screening of donations for the presence of *Treponema pallidum* (syphilis). Other non-viral infectious agents are also potentially transmissible by blood transfusion. However there are some infectious agents for which there are no routine tests available to prevent the disease from being transmitted by transfusion. For example, Trypanosoma cruzi (Chagas Disease) and the bovine spongiform encephalopathy (BSE) causing prion responsible for variant Creutzfeldt-Jakob Disease. To minimise the risk of transmitting infectious agents to patients, all donors are subjected to stringent screening procedures.
Incidence	Variable.
Main clinical features	Features of specific clinical infection.
Investigation	Clinical assessment. Microbiological investigation.
Intervention	Treat specific diagnosis. Inform Lifeblood.

Appendix IV: Transfusion reactions continued

Variant Creutzfeldt-Jakob Disease (vCJD)

Usual aetiology	A prion disease affecting the central nervous system due to consumption of beef products from cattle infected with the BSE-causing prion.
Incidence	To date, there have been no reported cases of vCJD in Australia. In the UK, there have been a small number of reported cases of putative transfusion transmission since 2004. There have been no reported cases of transmission by transfusion of sporadic Creutzfeldt-Jakob Disease (sCJD), and retrospective studies suggest that the possibility of such transmission of sCJD is remote.[31]
Main clinical features	Patients with vCJD frequently show prominent sensory disturbances and psychiatric symptoms with a minority having psychosis. A small proportion initially present with neurologic symptoms, including an early onset of sensory abnormalities.
Investigation	There are currently no routinely available tests to predict or prevent vCJD. Clinically assess for neurodegenerative signs and symptoms. MRI is the most useful non-invasive test for a positive diagnosis. Immediately consult with experts.
Intervention	Seek expert advice. Inform Lifeblood.



Appendix V: Residual risk estimates for transfusion-transmitted infections

Australia has one of the safest blood supplies in the world in terms of viral safety.

Lifeblood publishes estimates of the residual risks of transfusion-transmissible infection as a guide for clinicians in transfusion decision-making and informed consent processes (see Table 1).

These estimates have been derived using published models and are updated periodically.

The relative risks of transfusion-transmitted infections are very small when compared to other health-related risks (see Table 2).

Table 1: Residual risk estimates for transfusion-transmitted infections

Agent and testing standard	Window period	Estimate of residual risk 'per unit'[(a)(b)]
HIV (antibody/p24Ag + NAT)	5 days	Less than 1 in 1 million[32]
HCV (antibody + NAT)	3 days	Less than 1 in 1 million[32]
HBV (HBsAg + NAT)	17 days	Less than 1 in 1 million[32,33]
HTLV-1 and HTLV-2 (antibody)	51 days	Less than 1 in 1 million[34]
vCJD (No testing)		Less than 1 in 1 million[35]
Malaria (*Plasmodium* antibody)	7–14 days	Less than 1 in 1 million[30]
Syphilis (*T. pallidum* antibody)	30 days	Less than 1 in 1 million[36]

Table 2: General transfusion-related risks

Magnitude of risk	Risks of transfusion
Less than 1:1,000,000	Septic reaction from red cells Transfusion-transmitted Human Immunodeficiency Virus (HIV) Transfusion-transmitted Hepatitis B Virus (HBV) Transfusion-transmitted Hepatitis C Virus (HCV)
1:250,000	Septic reaction from platelets
1:190,000	Transfusion-related acute lung injury (TRALI)
1:50,000	Anaphylaxis
1:1000	Febrile non-haemolytic transfusion reaction (FNHTR)

Notes:

(a) HIV, HCV and HBV risk estimates are based on Lifeblood data from 1 January 2022 to 31 December 2023 (or with an earlier start date where required).

(b) The HTLV risk estimate is based on testing data from 1 January 2011 to 6 December 2020 and donor/donation data from 1 January 2022 to 31 December 2023.

References

1. National Blood Authority. Patient Blood Management Guidelines. Canberra: NBA (Available at: https://www.blood.gov.au/clinical-guidance/patient-blood-management#pbm-guidelines).
2. Australian and New Zealand Society of Blood Transfusion. Guidelines for Transfusion and Immunohaematology Laboratory Practice, Revised 1st edn. Sydney: ANZSBT Ltd, January 2020. (Available at: https://anzsbt.org.au/guidelines-standards/anzsbt-guidelines/).
3. Australian and New Zealand Society of Blood Transfusion and Australian College of Nursing. Guidelines for the Administration of Blood Products. 3rd edn. Sydney: ANZSBT Ltd, Minor revision February 2024 (Available at: https://anzsbt.org.au/guidelines-standards/anzsbt-guidelines/).
4. Standards Australia. AS 3864.2-2012 Medical Refrigeration Equipment - For The Storage Of Blood And Blood Products Part 2: User Related Requirements for Care, Maintenance, Performance Verification and Calibration. Sydney: Intertek Inform, 2012.
5. Australian Commission on Safety and Quality in Health Care (ACSQHC). Blood Management Standard. In: National Safety And Quality Health Service Standards. Sydney: ACSQHC, May 2021. (Available at: https://www.safetyandquality.gov.au/standards/nsqhs-standards).
6. Australian Red Cross Lifeblood. Blood Book: Australian Blood Administration Handbook. Revised 1st edn. June 2020 (Available at: https://www.lifeblood.com.au/sites/default/files/resource-library/2023-08/Blood-Book-ABAH_FINAL-v1.2-AUGUST-2023-SCREEN-WITH-LETTER.pdf).
7. Australian Commission on Safety and Quality in Health Care. Requirements for transfusion laboratory practice. 5th ed. Sydney: ACSQHC; 2022 (Available at: https://www.safetyandquality.gov.au/publications-and-resources/resource-library/requirements-transfusion-laboratory-practice-fifth-edition).
8. World Health Organization (WHO). Educational Modules on Clinical Use of Blood. Geneva: WHO Blood Transfusion Safety, 2021. (Available at: https://www.who.int/publications/i/item/9789240033733).
9. British Society for Haematology. Guideline on the Administration of Blood Components. London: British Society of Haematology, 6 November 2017. (Available at: https://b-s-h.org.uk/guidelines/guidelines/administration-of-blood-components).
10. McQuilten ZK, Wood EM, Yazer MH, Murphy MF. Chapter 25 - Blood Transfusion in Hospitals. In: Murphy MF, Roberts DJ, Yazer MH, Dunbar NM eds. Practical Transfusion Medicine. 6th edn. Oxford: Wiley, 2022. https://doi.org/10.1002/9781119665885.ch25
11. European Directorate for the Quality of Medicines and Healthcare of the Council of Europe (EDQM). Guide to the Preparation, Use and Quality Assurance of Blood Components. 19th edn. Strasbourg: EQQM Publications, 2017.
12. British Society for Haematology. Guideline for the Use of Anti-D Immunoglobulin for the Prevention of Haemolytic Disease of the Fetus and Newborn. London: British Society of Haematology, 21 January 2014. (Available at: https://b-s-h.org.uk/guidelines/guidelines/use-of-anti-d-immunoglobulin-for-the-prevention-of-haemolytic-disease-of-the-fetus-and-newborn).
13. Robinson D, McFadyen J, Merriman E, Tan CW, Baker R, Tran H. Updated recommendations for warfarin reversal in the setting of four-factor prothrombin complex concentrate. Med J Aust 2025; 222(1): 49-51. (Available at: https://doi.org/10.5694/mja2.52538).
14. AABB, American Red Cross, America's Blood Centers, and the Armed Services Blood Program. Circular of Information for the Use of Human Blood and Blood Components. Bethesda: AABB, December 2021. (Available at: https://www.aabb.org/news-resources/resources/circular-of-information).
15. Downes KA, Wilson E, Yomtovian R, Sarode R. Serial Measurement of Clotting Factors in Thawed Plasma Stored for 5 Days. Transfusion 2001; 41(4): 570. (Available at: https://onlinelibrary.wiley.com/doi/10.1046/j.1537-2995.2001.41040570.x).
16. Sheffield WP, Bhakta V, Mastronardi C, Ramirez-Arcos S, Howe D, Jenkins C. Changes in Coagulation Factor Activity and Content of Di (2-Ethylhexyl) Phthalate in Frozen Plasma Units During Refrigerated Storage for Up to Five Days after Thawing. Transfusion 2012; 52(3): 493–502. (Available at: https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1537-2995.2011.03300.x).
17. Australian and New Zealand Society of Blood Transfusion. Thawed Plasma Components: A Framework for Preparation, Storage and Use. 2nd edn. Sydney: ANZSBT Ltd, 2013. (Available at: https://anzsbt.org.au/guidelines-standards/anzsbt-guidelines/).
18. Reference Standard 5.1.8A – Requirements for Storage, Transportation and Expiration. In: AABB. Standards for Blood Banks and Transfusion Services. 32nd edn. Bethesda: AABB, 2020.
19. Bowden RA, Slichter SJ, Sayers M, Weisdorf D, Cays M, Schoch G, et al. A Comparison of Filtered Leucocyte-Reduced and Cytomegalovirus Seronegative Blood Products for the Prevention of Transfusion-Associated CMV Infection after Marrow Transplant. Blood 1995; 86: 3598–3603. (Available at: https://doi.org/10.1182/blood.V86.9.3598.bloodjournal8693598).

References continued

20. Nichols WG, Price TH, Gooley T, Corey L, Boeckh M. Transfusion-Transmitted Cytomegalovirus Infection after Receipt of Leukoreduced Blood Products. Blood 2003; 101: 4195–4200. (Available at: https://doi.org/10.1182/blood-2002-10-3143).
21. Vamvakas EC. Is White Blood Cell Reduction Equivalent to Antibody Screening in Preventing Transmission of Cytomegalovirus by Transfusion? A Review of the Literature and Meta-analysis. Transfusion Medicine Reviews 2005; 19(3): 185–199. (Available at: https://www.sciencedirect.com/science/article/pii/S088779630500012X?via%3Dihub).
22. Wu Y, et al. Direct Assessment Of Cytomegalovirus Transfusion Transmitted Risks After Universal Leukoreduction. Transfusion 2010; 50: 776–786. (Available at: https://doi.org/10.1111/j.1537-2995.2009.02486.x).
23. Australian and New Zealand Society of Blood Transfusion. Guidelines for Prevention of Transfusion-Associated Graft-Versus-Host Disease (TA-GvHD). 2nd edn. Sydney: ANZSBT Ltd, January 2024. (Available at: https://anzsbt.org.au/guidelines-standards/anzsbt-guidelines/).
24. Kopko PM, ed. Transfusion Reactions, 5th edn. Bethesda: AABB Press, 2021.
25. Crowder LA, Saá P. Chapter 7 – Infectious Disease Screening. In: Cohn CS, Delaney M, Johnson ST, Katz LM, Schwartz J(Y) eds. AABB Technical Manual. 21st edn. Bethesda: AABB, 2023.
26. Francis RO, Stotler BA. Chapter 22 – Noninfectious Complications of Blood Transfusion. In: Cohn CS, Delaney M, Johnson ST, Katz LM, Schwartz J(Y) eds. AABB Technical Manual. 21st edn. Bethesda: AABB, 2023.
27. Eder A, Kennedy JM, Dy BA, Notari EP, Weiss JW, Fang CT, et al. Bacterial Screening of Apheresis Platelets and the Residual Risk of Septic Transfusion Reactions: The American Red Cross Experience (2004–2006). Transfusion 2007; 47(7): 1134–1142. (Available at: https://doi.org/10.1111/j.1537-2995.2007.01248.x).
28. Kuehnert MJ, Roth VR, Haley NR, Gregory KR, Elder KV, Schreiber GB, et al. Transfusion-Transmitted Bacterial Infection in the United States, 1998 through 2000. Transfusion 2001; 41(12): 1493–1499. (Available at: https://doi.org/10.1046/j.1537-2995.2001.41121493.x).
29. Cheng A, Das A, Chaw K, Dennington PM, Styles CE, Gosbell IB. Safety analysis of extended platelet shelf-life with large-volume delayed sampling on BACT/ALERT VIRTUO in Australia. Microorganisms. 2023; 11: 2346. (Available at: https://doi.org/10.3390/microorganisms11092346)
30. Seed CR, Kee G, Wong T, Law M, Ismay S. Assessing the Safety and Efficacy of a Test Based, Targeted Screening Strategy to Minimise Transfusion Transmitted Malaria. Vox Sang 2010; 98:e182–192. (32). (Available at: https://doi.org/10.1111/j.1423-0410.2009.01251.x).
31. Crowder LA, Dodd RY, Schonberger LB. Absence of evidence of transfusion transmission risk of Creutzfeldt-Jakob disease in the United States: Results from a 28-year lookback study. Transfusion. 2024; 64(6): 980-985 (Available at: https://doi.org/10.1111/trf.17837).
32. Weusten J, Vermeulen M, van Drimmelen H, et al. Refinement Of A Viral Transmission Risk Model for Blood Donations in Seroconversion Window Phase Screened by Nucleic Acid Testing in Different Pool Sizes and Repeat Test Algorithms. Transfusion 2011; 51: 203-215. (Available at: https://doi.org/10.1111/j.1537-2995.2010.02804.x).
33. Seed CR, Kiely P, Hoad VC, Keller AJ: Refining the Risk Estimate for Transfusion-Transmission of Occult Hepatitis B Virus. Vox Sang 2017; 112(1): 3-8. (Available at: https://doi.org/10.1111/vox.12446).
34. Styles CE, Seed CR, Hoad VC, et al. Reconsideration of blood donation testing strategy for human T-cell lymphotropic virus in Australia. Vox Sang. 2017; 112:723-732. (Available at: https://doi.org/10.1111/vox.12597).
35. McManus, H, Seed, CR, Hoad, VC, Kiely, P, Kaldor, JM, Styles, CE, et al. Risk of variant Creutzfeldt–Jakob disease transmission by blood transfusion in Australia. Vox Sang 2022; 117(8): 1016-1026. (Available at: https://doi.org/10.1111/vox.13290).
36. Jayawardena T, Hoad V, Styles C, et al. Modelling the risk of transfusion-transmitted syphilis: a reconsideration of blood donation testing strategies. Vox Sang 2019; 114(2): 107-116. (Available at: https://doi.org/10.1111/vox.12741).

Australian Red Cross Lifeblood

National Office

100–154 Batman Street
West Melbourne VIC 3003

E: transfusion@redcrossblood.org.au

Customer contacts

We provide general customer service and specialist medical support 24-hours a day for health professionals.

For our full list of contact details visit lifeblood.com.au/contact

© Copyright 2025 Australian Red Cross Lifeblood

The information contained in this booklet was correct at the time of publishing: March 2025

No person should act on the basis of the contents of this publication without first obtaining specific, independent and relevant advice.

Australian Red Cross Lifeblood is not liable for any loss, damage, cost or expense incurred or arising by any person, or organisation, using or relying on the information in this publication. All images and photographs in this book are representative only. Actual component labels and product packaging may vary. All rights reserved.

Apart from any fair dealing for the use of private study, research, criticism, or review as permitted under the Copyright Act, no part of this book may be transmitted or reproduced in any form, electronic or mechanical, or by any information storage and retrieval system, without the written permission of the publishers.

Australian governments fund Australian Red Cross Lifeblood to provide blood, blood products and services to the Australian community.

To give blood, visit lifeblood.com.au or call **13 14 95**